Dorey's Resource Book

Topic 2 – Integral Calculus

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1 The definite integral from geometry

ANCHOR QUESTION

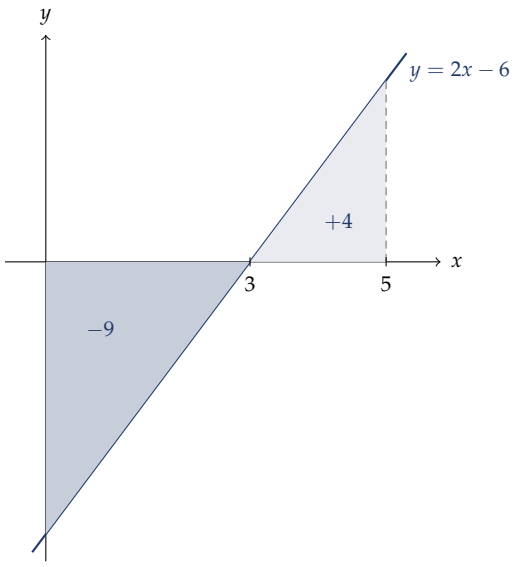
Evaluate $\int_0^5 (2x-6)\,dx$ *by finding areas.*

SYLLABUS MAV-12-05

- Examine for a function $f(x)$ which indicates the rate of change of a quantity, the meaning of $\sum_a^b f(x)\,\Delta x$ as an estimate of the total change over $a \le x \le b$
- Consider the definite integral as $\int_a^b f(x)\,dx = \lim_{\Delta x \to 0} \sum_a^b f(x)\,\Delta x$, noting that the result will be negative when $f(x) \le 0$ throughout the interval
- Evaluate the definite integral $\int_a^b f(x)\,dx$ by calculating areas using geometrical formulas, where $f(x) \ge 0$, $f(x) \le 0$ or $f(x)$ changes sign in $a \le x \le b$

NOTES

- $\sum_a^b f(x)\,\Delta x$ adds the areas of thin rectangles of height $f(x)$ and width Δx.
- $\int_a^b f(x)\,dx = \lim_{\Delta x \to 0} \sum_a^b f(x)\,\Delta x$: the sum as the rectangles get thinner.
- The integral is a *signed* area: area above the x-axis counts positive and area below counts negative.
- Triangle $\frac{1}{2}bh$, trapezium $\frac{1}{2}h(a+b)$, circle πr^2. $y = \sqrt{r^2 - x^2}$ is the upper semicircle of radius r.
- Find where the graph crosses the axis, split there, find each area, and give each the sign of its side.



WORKED EXAMPLE

Evaluate $\int_1^6 (2x-4)\,dx$ *using areas.*

The line $y = 2x - 4$ crosses the x-axis at $x = 2$. where the sign changes
From 1 to 2: triangle below the axis, base 1, height 2: area 1, so -1.
From 2 to 6: triangle above the axis, base 4, height 8: area 16.
$\int_1^6 (2x-4)\,dx = -1 + 16 = 15$

Tier 1 · sums of $f(x)\Delta x$

1. Let $f(x) = x + 2$ on $0 \le x \le 4$ with $\Delta x = 1$.

(a) Find the left sum $f(0)\Delta x + f(1)\Delta x + f(2)\Delta x + f(3)\Delta x$.

(b) Find the right sum $f(1)\Delta x + \cdots + f(4)\Delta x$.

(c) Find $\int_0^4 (x+2)\,dx$ using a trapezium, and compare.

2. Let $f(x) = x^2$ on $0 \le x \le 3$ with $\Delta x = 1$. Find the left sum and the right sum.

3. Let $f(x) = 2x$ on $0 \le x \le 2$ with $\Delta x = 0.5$.

(a) Find the left sum and the right sum.

(b) Use a triangle to find $\int_0^2 2x\,dx$.

Tier 2 · areas above the axis

4. Sketch $y = 3$ and find $\int_1^5 3\,dx$.

5. Find $\int_0^4 \frac{x}{2}\,dx$ and $\int_0^6 (x+1)\,dx$ using areas.

6. Evaluate $\int_2^5 (2x-3)\,dx$ using a trapezium.

Tier 3 · areas below the axis

7. Evaluate $\int_0^4 (-2)\,dx$.

8. Evaluate $\int_0^3 (x-3)\,dx$ using a triangle.

9. Evaluate $\int_1^4 (-x)\,dx$ using a trapezium.

Tier 4 · the graph crosses the axis

10. Evaluate $\displaystyle\int_0^6 (x-2)\,dx$ using areas.

11. Consider $\displaystyle\int_{-1}^5 (2x-4)\,dx$.

(a) Evaluate the integral using areas.

(b) Find the total area between the line and the x-axis.

12. Evaluate $\displaystyle\int_0^7 (4-x)\,dx$ and find the total area between the line and the x-axis.

Tier 5 · semicircles and combined shapes

13. Evaluate $\displaystyle\int_{-2}^2 \sqrt{4-x^2}\,dx$.

14. Evaluate $\displaystyle\int_{-5}^5 \left(-\sqrt{25-x^2}\right)dx$.

15. Evaluate $\displaystyle\int_{-3}^3 \left(2+\sqrt{9-x^2}\right)dx$.

Conceptual

16. Explain why $\displaystyle\int_a^b f(x)\,dx$ can be negative, but an area cannot.

17. Is it true that if $\displaystyle\int_0^4 f(x)\,dx = 0$ then $f(x) = 0$ for all x in $[0,4]$? Give an example.

18. Use symmetry to explain why $\displaystyle\int_{-3}^3 x\,dx = 0$.

19. What happens to the left and right sums of an increasing function as $\Delta x \to 0$?

20. The function f is defined by $f(x) = 2$ for $0 \le x \le 2$ and $f(x) = 4 - x$ for $2 \le x \le 6$.

(a) Find $\displaystyle\int_0^2 f(x)\,dx$. (1 mark)

(b) Find $\displaystyle\int_2^6 f(x)\,dx$. (2 marks)

(c) Find the total area between the graph of f and the x-axis for $0 \le x \le 6$. (2 marks)

Revision

21. Let $f(x) = 2x^2 - 8x + k$, where k is a constant.

(a) Find the equation of the axis of symmetry of $y = f(x)$. (1 mark)

(b) The equation $f(x) = 0$ has exactly one real root. Find k. (2 marks)

(c) When $k = 6$, solve $f(x) = 0$. (2 marks)

22. A sector OAB has radius 6 cm and angle $\frac{\pi}{3}$ at the centre O.

(a) Find the length of the arc AB. (1 mark)

(b) Find the area of the sector. (2 marks)

(c) Find the exact area of the segment between the chord AB and the arc. (2 marks)

23. Let $f(x) = xe^{2x}$.

(a) Show that $f'(x) = e^{2x}(1 + 2x)$. (2 marks)

(b) Find the coordinates of the stationary point. (1 mark)

(c) Determine its nature. (2 marks)

2 Primitives of x^n and $(ax+b)^n$

ANCHOR QUESTION

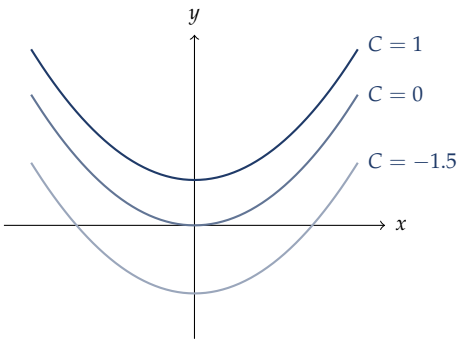
Find the primitive of $9x^2 - 4x + 7$.

SYLLABUS MAV-12-05

- Define a primitive of a function $f(x)$ as a function $F(x)$ whose derivative $F'(x) = f(x)$ and recognise the process of finding the primitive as the reverse of differentiation
- Recognise that a function whose derivative is everywhere zero is a constant function
- Prove by differentiation that a primitive of $f(x) = x^n$ is $F(x) = \dfrac{x^{n+1}}{n+1}$, for all real $n \neq -1$
- Prove by differentiation that if $F(x)$ and $G(x)$ are primitives of $f(x)$ and $g(x)$, and k is a constant, then $F(x) + G(x)$ is a primitive of $f(x) + g(x)$, and $kF(x)$ is a primitive of $kf(x)$
- Recognise that primitives are not unique, and that any two primitives of $f(x)$ differ by a constant, so that the general primitive is $F(x) + C$
- Determine the primitive of a sum of functions of the form kx^n for all real $n \neq -1$, and of $(ax+b)^n$
- Use algebraic manipulation to express given functions in forms suitable for determining primitive functions
- Determine $f(x)$, given $f'(x)$ and an initial condition $f(a) = b$

NOTES

- A primitive of $f(x)$ is a function $F(x)$ with $F'(x) = f(x)$. Integration reverses differentiation.
- Primitives are not unique: any two differ by a constant, so the general primitive is $F(x) + C$.
- $\displaystyle\int x^n\,dx = \frac{x^{n+1}}{n+1} + C$ for $n \neq -1$: add 1 to the power, then divide by the new power.
- Integrate term by term, and constants multiply through: $\int \big(f(x) + g(x)\big)dx = \int f\,dx + \int g\,dx$ and $\int kf\,dx = k\int f\,dx$.
- Write roots, reciprocals and brackets as powers of x before integrating: $\sqrt{x} = x^{\frac{1}{2}}$ and $\dfrac{1}{x^2} = x^{-2}$. Expand products and divide terms by the denominator.
- $\displaystyle\int (ax+b)^n\,dx = \frac{(ax+b)^{n+1}}{a(n+1)} + C$ for $n \neq -1$: also divide by a.
- To find $f(x)$ from $f'(x)$ and $f(a) = b$: integrate, then substitute $x = a$ to find C.



WORKED EXAMPLE

Find the primitive of $6x^2 + 10x - 3$.

$$\frac{6x^3}{3} + \frac{10x^2}{2} - 3x + C$$
$$= 2x^3 + 5x^2 - 3x + C$$

Tier 1 · powers of x

24. Find the primitive of each.

(a) x^5

(b) $4x^3$

(c) $10x^9$

(d) 7

25. Show by differentiation that $\frac{x^6}{6}$ is a primitive of x^5, and write the general primitive.

26. Find $\int 6x^2\,dx$, $\int x\,dx$ and $\int 1\,dx$.

Tier 2 · sums and constants

27. Find the primitive of $12x^3 + 6x - 7$.

28. Find $\int \left(x^4 + 2x^3 - x\right) dx$.

29. Find $\int \left(\frac{x^2}{3} - 5x + 2\right) dx$.

Tier 3 · negative and fractional powers

30. Find $\int \left(x^3 - \frac{4}{x^3}\right) dx$.

31. Find $\int \left(\sqrt{x} + \frac{1}{\sqrt{x}}\right) dx$.

32. Find $\int \frac{1}{x^2}\,dx$ and $\int \frac{3}{\sqrt[3]{x}}\,dx$.

Tier 4 · algebra first

33. Find the primitive of $(x+3)(2x-5)$.

34. Find $\displaystyle\int \frac{x^4-3x^2+2}{x^2}\,dx$.

35. Find $\displaystyle\int \left(\sqrt{x}+2\right)^2\,dx$.

Tier 5 · $(ax+b)^n$

36. Find the primitive of each.

(a) $(2x+1)^3$

(b) $(3x-2)^5$

(c) $\dfrac{1}{(x+4)^2}$

37. Find $\displaystyle\int \sqrt{4x+1}\,dx$.

38. Find $\displaystyle\int \frac{6}{(3x-1)^2}\,dx$.

Tier 6 · initial conditions

39. Given $f'(x)=6x^2-2$ and $f(1)=5$, find $f(x)$.

40. Given $f'(x)=3\sqrt{x}$ and $f(4)=10$, find $f(x)$.

41. A curve has gradient $\dfrac{dy}{dx}=8(2x-1)^3$ and passes through $(1,3)$. Find its equation.

Conceptual

42. Prove by differentiation that a primitive of x^n is $\dfrac{x^{n+1}}{n+1}$ for $n \neq -1$.

43. Explain what goes wrong if you apply the rule $\dfrac{x^{n+1}}{n+1}$ to x^{-1}.

44. Show that $(x^2+1)^2$ and x^4+2x^2 are both primitives of $4x^3+4x$, and explain how they are related.

45. A curve has $\dfrac{dy}{dx} = 6x^2 - 4x$ and passes through $(1,3)$.

(a) Find the equation of the curve. (2 marks)

(b) Show that the curve has stationary points at $x = 0$ and $x = \frac{2}{3}$, and find their y-coordinates. (2 marks)

46. What is $\displaystyle\int (3x-1)^4\,dx$?

A. $\dfrac{(3x-1)^5}{5} + C$ B. $\dfrac{(3x-1)^5}{15} + C$ C. $12(3x-1)^3 + C$ D. $\dfrac{(3x-1)^5}{3} + C$

Revision

47. An arithmetic sequence has $T_3 = 11$ and $T_8 = 26$.

(a) Find the common difference. (1 mark)

(b) Find the first term. (1 mark)

(c) Find the sum of the first 20 terms. (2 marks)

48. Let $f(x) = 2x - 4$ for $0 \leq x \leq 6$.

(a) Find the x-intercept of the graph. (1 mark)

(b) Use areas to evaluate $\displaystyle\int_0^6 (2x-4)\,dx$. (2 marks)

(c) Find the total area between the line and the x-axis. (1 mark)

49. Consider the equation $2^{2x} - 5 \times 2^x + 4 = 0$.

(a) Show that $2^x = 1$ or $2^x = 4$. (2 marks)

(b) Hence solve the equation. (1 mark)

(c) Evaluate $\log_3 54 - \log_3 2$. (1 mark)

3 The Fundamental Theorem of Calculus

ANCHOR QUESTION

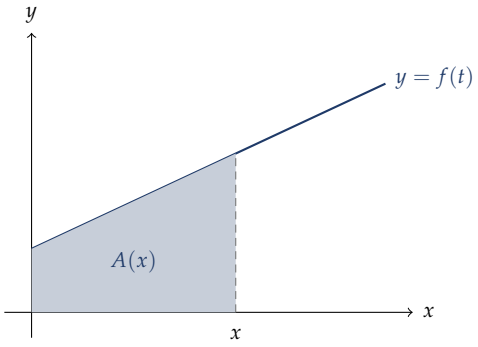
Evaluate $\int_1^3 (4x^3 - 2x)\,dx$.

SYLLABUS MAV-12-05

- Consider the function defined by $A(x) = \int_a^x f(t)\,dt$ and recognise that $A(x)$ is a primitive of $f(x)$
- Recognise the Fundamental Theorem of Calculus as $\int_a^b f(x)\,dx = \Big[F(x)\Big]_a^b = F(b) - F(a)$ for a continuous function f where $F(x)$ is any primitive of $f(x)$
- Use the notation $\int f(x)\,dx$ for the general primitive of $f(x)$, called the indefinite integral, so $\int f(x)\,dx = F(x) + C$
- Use $\int_a^b f(x)\,dx + \int_b^c f(x)\,dx = \int_a^c f(x)\,dx$ and $\int_a^b f(x)\,dx = -\int_b^a f(x)\,dx$
- Recognise and use symmetry, particularly odd and even functions, to simplify and solve integration problems

NOTES

- $\int f(x)\,dx = F(x) + C$ is the indefinite integral: the general primitive.
- Fundamental Theorem: $\int_a^b f(x)\,dx = F(b) - F(a)$, where F is any primitive of f. Write $\Big[F(x)\Big]_a^b$ first.
- No $+C$ in a definite integral: it cancels in $F(b) - F(a)$.
- $A(x) = \int_a^x f(t)\,dt$ is the area gathered up to x. Then $A'(x) = f(x)$.
- $\int_a^b + \int_b^c = \int_a^c$ and swapping the limits changes the sign.
- Odd function ($f(-x) = -f(x)$): $\int_{-a}^a f(x)\,dx = 0$. Even function ($f(-x) = f(x)$): $\int_{-a}^a f(x)\,dx = 2\int_0^a f(x)\,dx$.



WORKED EXAMPLE

Evaluate $\displaystyle\int_{-1}^{2} (3x^2 - 4x + 1)\,dx$.

$$\int_{-1}^{2} (3x^2 - 4x + 1)\,dx = \Big[x^3 - 2x^2 + x\Big]_{-1}^{2}$$
$$= (8 - 8 + 2) - (-1 - 2 - 1) = 2 + 4 = 6$$

Tier 1 · indefinite and definite integrals

50. Find $\displaystyle\int (6x^2 - 2x)\,dx$ and then $\displaystyle\int_1^2 (6x^2 - 2x)\,dx$.

51. Evaluate each.

(a) $\displaystyle\int_0^3 x^2\,dx$

(b) $\displaystyle\int_1^2 x^3\,dx$

(c) $\displaystyle\int_0^5 4\,dx$

52. Explain why no $+C$ is needed in a definite integral.

Tier 2 · polynomials

53. Evaluate $\displaystyle\int_0^2 (3x^2 + 1)\,dx$.

54. Evaluate $\displaystyle\int_1^3 (4x^3 - 2x)\,dx$.

55. Evaluate $\displaystyle\int_{-2}^1 (x^2 - x + 3)\,dx$.

Tier 3 · negative and fractional powers

56. Evaluate $\displaystyle\int_1^4 \sqrt{x}\,dx$.

57. Evaluate $\displaystyle\int_1^2 \frac{1}{x^2}\,dx$.

58. Evaluate $\displaystyle\int_1^4 \left(x^{-\frac{1}{2}} + x\right)dx$.

Tier 4 · $(ax+b)^n$ with limits

59. Evaluate $\displaystyle\int_0^1 (2x+1)^3\,dx$.

60. Evaluate $\displaystyle\int_1^3 \frac{1}{(x+1)^2}\,dx$.

61. Show that $\displaystyle\int_0^2 \sqrt{2x+1}\,dx = \frac{5\sqrt{5}-1}{3}$.

Tier 5 · limits and the properties of integrals

62. Given $\displaystyle\int_0^3 f(x)\,dx = 7$ and $\displaystyle\int_0^5 f(x)\,dx = 12$, find:

(a) $\displaystyle\int_3^5 f(x)\,dx$

(b) $\displaystyle\int_5^0 f(x)\,dx$

(c) $\displaystyle\int_0^3 \big(2f(x)+1\big)dx$

63. Show that $\displaystyle\int_2^2 \left(x^3+1\right)dx = 0$, and that $\displaystyle\int_1^3 x^2\,dx = -\int_3^1 x^2\,dx$.

64. Evaluate $\displaystyle\int_0^2 x^2\,dx + \int_2^4 x^2\,dx$ by combining into one integral.

Tier 6 · odd and even functions

65. Evaluate $\displaystyle\int_{-2}^2 \left(x^3+x\right)dx$.

66. Evaluate $\displaystyle\int_{-2}^2 x^4\,dx$ using $2\displaystyle\int_0^2 x^4\,dx$.

67. Evaluate $\displaystyle\int_{-1}^1 \left(x^5+3x^2\right)dx$.

Tier 7 · the function $A(x)$

68. Let $A(x) = \int_1^x (3t^2 - 2)\,dt$.

(a) Find $A(x)$.

(b) Find $A'(x)$ and $A(1)$.

69. Let $A(x) = \int_0^x (4t - t^2)\,dt$. Find $A(x)$ and find where $A'(x) = 0$.

70. Find $\dfrac{d}{dx}\displaystyle\int_2^x \sqrt{t^2+1}\,dt$.

Conceptual

71. Explain why $A'(x) = f(x)$ when $A(x) = \displaystyle\int_a^x f(t)\,dt$.

72. Find a if $\displaystyle\int_0^a (2x+1)\,dx = 6$, where $a > 0$.

73. Show that $\displaystyle\int_{-a}^{a} x^5\,dx = 0$ for every a, and explain using odd functions.

74. Let $A(x) = \displaystyle\int_0^x f(t)\,dt$, where $f(t) = t(t-3)$.

(a) Find $A(x)$. (2 marks)

(b) Find the values of $x > 0$ where $A(x) = 0$. (2 marks)

(c) Explain why $A(x)$ has a minimum at $x = 3$. (1 mark)

75. What is $\dfrac{d}{dx}\displaystyle\int_1^x \sqrt{t^2+1}\,dt$?

A. $\sqrt{x^2+1}$ B. $\dfrac{x}{\sqrt{x^2+1}}$ C. $\sqrt{2}$ D. $\dfrac{x\sqrt{x^2+1}}{3}$

Revision

76. Let $f(x) = x^3 - 3x^2 + k$, where k is a constant.

(a) Find the x-coordinates of the stationary points. (2 marks)

(b) Determine the nature of each. (2 marks)

(c) Find k if the minimum value is -1. (1 mark)

77. A bag holds 4 red and 3 blue counters. Two counters are drawn without replacement.

(a) Find the probability both are red. (1 mark)

(b) Find the probability they are different colours. (2 marks)

(c) Given that they are different colours, find the probability the first was red. (2 marks)

78. Solve the following for $0 \leq x \leq 2\pi$.

(a) $2\sin x = \sqrt{3}$ (2 marks)

(b) $2\cos^2 x = \cos x$ (3 marks)

4 The reverse chain rule

ANCHOR QUESTION

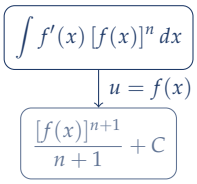
Find $\int 6x\left(3x^2+1\right)^4\,dx$.

SYLLABUS MAV-12-05

- Prove by differentiation, and apply $\int u^n \frac{du}{dx}\,dx = \frac{1}{n+1}u^{n+1}+C$, where u is a function of x, or $\int f'(x)\left[f(x)\right]^n dx = \frac{1}{n+1}\left[f(x)\right]^{n+1}+C$, for real $n \neq -1$

NOTES

- $\int f'(x)\left[f(x)\right]^n dx = \frac{\left[f(x)\right]^{n+1}}{n+1}+C$ for $n \neq -1$.
- Proof: $\frac{d}{dx}\frac{[f(x)]^{n+1}}{n+1} = [f(x)]^n \times f'(x)$ by the chain rule.
- Look for a bracket (the inside f) and, outside it, the derivative f' of that bracket.
- If the factor outside is a constant multiple k of f', write kf' and divide the answer by k.
- A bracket in a denominator or a root is a power: $\frac{1}{u^2} = u^{-2}$ and $\sqrt{u} = u^{\frac{1}{2}}$.
- The factor outside the bracket must match f' exactly (up to a constant). Otherwise this method does not apply.



WORKED EXAMPLE

Find $\int 4x\left(2x^2-3\right)^5\,dx$.

$f(x) = 2x^2-3$ and $f'(x) = 4x$, which is exactly the factor outside. inside and its derivative

$\int f'(x)\left[f(x)\right]^5 dx = \frac{\left(2x^2-3\right)^6}{6}+C$

Tier 1 · find f, f' and k

79. For each, write the bracket $f(x)$, its derivative $f'(x)$ and the constant k so that the integrand is $k\,f'(x)\,[f(x)]^n$.

(a) $x\left(x^2+4\right)^3$

(b) $6x^2\left(x^3-1\right)^4$

(c) $(2x+3)\left(x^2+3x\right)^2$

80. Which of these has the form $k\,f'(x)\,[f(x)]^n$? Give f for those that do.

(a) $x^2\left(x^3+5\right)^2$

(b) $x\left(x^3+5\right)^2$

(c) $\left(x^2+1\right)^3$

81. Find $f'(x)$ for $f(x)=5x^3-2x$, and write $15x^2-2$ in terms of f'.

Tier 2 · the derivative is exactly there

82. Find $\displaystyle\int 2x\left(x^2+1\right)^4\,dx$.

83. Find $\displaystyle\int 3x^2\left(x^3-2\right)^3\,dx$.

84. Find $\displaystyle\int (2x+3)\left(x^2+3x\right)^2\,dx$.

Tier 3 · a constant to adjust

85. Find $\displaystyle\int x\left(x^2+4\right)^3\,dx$.

86. Find $\displaystyle\int x^2\left(x^3+1\right)^2\,dx$.

87. Find $\displaystyle\int x\left(1-x^2\right)^5\,dx$.

Tier 4 · roots and reciprocals

88. Find $\displaystyle\int \frac{x}{\sqrt{x^2+9}}\,dx$.

89. Find $\displaystyle\int \frac{x}{\left(x^2+1\right)^2}\,dx$.

90. Find $\displaystyle\int x^2\sqrt{x^3+1}\,dx$.

Tier 5 · definite integrals

91. Evaluate $\displaystyle\int_0^1 x\left(x^2+1\right)^3\,dx$.

92. Evaluate $\displaystyle\int_0^2 \frac{x}{\sqrt{x^2+5}}\,dx$.

93. Evaluate $\displaystyle\int_1^2 3x^2\left(x^3+2\right)^{-2}\,dx$ and give the answer as a fraction.

Tier 6 · unseen forms

94. Find $\displaystyle\int (x+1)\left(x^2+2x\right)^3\,dx$.

95. Find $\displaystyle\int \frac{2x+1}{\left(x^2+x\right)^3}\,dx$.

96. Find $\displaystyle\int \left(x^3-x\right)\left(x^4-2x^2\right)^5\,dx$.

Conceptual

97. Prove by differentiation that $\displaystyle\int f'(x)\left[f(x)\right]^n dx = \frac{\left[f(x)\right]^{n+1}}{n+1} + C$ for $n \neq -1$.

98. Explain why this method cannot be used for $\displaystyle\int \left(x^2+1\right)^4\,dx$.

99. A student writes $\displaystyle\int x\left(x^2+1\right)^3\,dx = \frac{x^2}{2} \times \frac{\left(x^2+1\right)^4}{4} + C$. Explain the error and correct it.

100. A curve has gradient $\dfrac{dy}{dx} = 6x\left(x^2-1\right)^2$ and passes through $(1,2)$.

(a) Find the equation of the curve. (2 marks)

(b) Find the stationary points and determine their nature. (3 marks)

101. What is $\displaystyle\int x\left(3x^2-2\right)^4\,dx$?

A. $\dfrac{\left(3x^2-2\right)^5}{30}+C$ B. $\dfrac{\left(3x^2-2\right)^5}{5}+C$ C. $\dfrac{\left(3x^2-2\right)^5}{15}+C$ D. $\dfrac{x^2\left(3x^2-2\right)^5}{10}+C$

Revision

102. Let $f(x) = \ln\left(x^2 + 1\right)$.

(a) Show that $f'(x) = \dfrac{2x}{x^2+1}$. (1 mark)

(b) Find the equation of the tangent at $x = 1$. (2 marks)

(c) Show that f is an even function. (1 mark)

103. Evaluate $\displaystyle\int_{-1}^{1} \left(x^3 + 2x^2 + 1\right) dx$.

(a) Identify the odd term and the even terms. (1 mark)

(b) Hence evaluate the integral. (2 marks)

104. A geometric sequence has first term 3 and common ratio 2.

(a) Find T_6. (1 mark)

(b) Find S_8. (2 marks)

(c) A different geometric series has first term 8 and ratio $\frac{1}{2}$. Find its limiting sum. (1 mark)

5 Integrating exponential functions

ANCHOR QUESTION

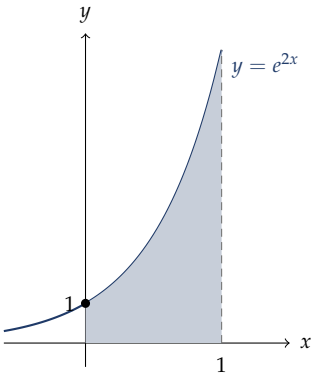
Find $\int \left(10e^{2x} - 3e^{-x}\right) dx$.

SYLLABUS MAV-12-05

- Establish and use the formula $\int e^x \, dx = e^x + C$
- Establish and use the formula $\int e^{ax+b} \, dx = \frac{1}{a}e^{ax+b} + C$, where a and b are constants and $a \neq 0$
- Establish and use the formula $\int a^x \, dx = \frac{a^x}{\ln a} + C$, where a is a constant and $a > 0$
- Establish and use $\int u'e^u \, dx = e^u + C$, where u is a function of x, or $\int f'(x)e^{f(x)}dx = e^{f(x)} + C$
- Find primitives of functions involving exponential functions

NOTES

- $\int e^x \, dx = e^x + C$. The power stays the same.
- $\int e^{ax+b} \, dx = \frac{1}{a}e^{ax+b} + C$ for $a \neq 0$: divide by the coefficient of x.
- $\int a^x \, dx = \frac{a^x}{\ln a} + C$ for $a > 0$, $a \neq 1$. Here $\ln a$ is just a number.
- $\int f'(x)\, e^{f(x)} \, dx = e^{f(x)} + C$. Adjust with a constant k when the factor is $kf'(x)$. This is on the reference sheet.
- Write $\frac{1}{e^{2x}}$ as e^{-2x} and expand products before integrating.
- e is a constant (≈ 2.718), so $\int e \, dx = ex$, not e^x.



WORKED EXAMPLE

Find $\int \left(8e^{4x} + 6e^{-3x}\right) dx.$

$$\int 8e^{4x}dx = \frac{8}{4}e^{4x} = 2e^{4x}$$ divide by $a = 4$

$$\int 6e^{-3x}dx = \frac{6}{-3}e^{-3x} = -2e^{-3x}$$

$$\int \left(8e^{4x} + 6e^{-3x}\right) dx = 2e^{4x} - 2e^{-3x} + C$$

Tier 1 · e^{ax}

105. Find each integral.

(a) $\int e^{4x}dx$

(b) $\int e^{-x}dx$

(c) $\int e^{\frac{x}{2}}dx$

106. Find each integral.

(a) $\int 8e^{-2x}dx$

(b) $\int \dfrac{e^{3x}}{2}\,dx$

107. Show by differentiation that $\dfrac{1}{5}e^{5x}$ is a primitive of e^{5x}.

Tier 2 · e^{ax+b} and sums

108. Find $\int 3e^{2x+5}dx.$

109. Find $\int \left(2e^{3x+1} - e^{-x}\right) dx.$

110. Find $\int 5e^{1-2x}dx.$

Tier 3 · a^x

111. Find each integral.

(a) $\int 2^x dx$

(b) $\int 5^x dx$

(c) $\int 3 \times 4^x dx$

112. Find $\int (3^x + x)\,dx$ and $\int (2^x - 2)\,dx$.

113. Find $\int \left(5^x - x^5\right) dx$.

Tier 4 · rewrite first

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114. Find $\int \dfrac{1}{e^{2x}}\,dx$.

115. Find $\int \left(e^x + e^{-x}\right)^2 dx$.

116. Find $\int \dfrac{e^{2x}+1}{e^x}\,dx$.

Tier 5 · definite integrals

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117. Evaluate $\int_0^1 \left(e^{2x} + 1\right) dx$.

118. Find the exact value of $\int_0^{\ln 2} e^{2x} dx$.

119. Find the exact value of $\int_{-1}^{1} e^{2x+1} dx$.

Tier 6 · $f'(x)e^{f(x)}$

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120. Find each integral.

(a) $\int 3x^2 e^{x^3} dx$

(b) $\int x e^{x^2} dx$

(c) $\int (2x-1)\, e^{x^2-x} dx$

121. Find $\int x^2 e^{x^3+1} dx$.

122. Evaluate $\int_0^1 (2x+1)\, e^{x^2+x} dx$.

123. Evaluate $\int_0^1 x e^{-x^2} dx$.

Conceptual

124. Prove that $\int e^{ax+b}dx = \frac{1}{a}e^{ax+b} + C$ for $a \neq 0$.

125. Why is $\int e\,dx = ex + C$ and not $e^x + C$?

126. What is $\int \left(e + e^{3x}\right) dx$, where e is a constant?

A. $ex + 3e^{3x} + c$ B. $ex + \frac{1}{3}e^{3x} + c$ C. $e + 3e^{3x} + c$ D. $e + \frac{1}{3}e^{3x} + c$

127. Given $f'(x) = 6e^{3x}$ and $f(0) = 5$, find $f(x)$.

128. Consider $f(x) = e^x + e^{-x}$.

(a) Show that $\int_0^{\ln 2} f(x)\,dx = \frac{3}{2}$. (2 marks)

(b) Explain why $f(x)$ is an even function and write $\int_{-\ln 2}^{\ln 2} f(x)\,dx$. (2 marks)

Revision

129. The gradient of a curve is $\frac{dy}{dx} = 3x^2 - 6x$ and the curve passes through $(1, 2)$.

(a) Find the equation of the curve. (2 marks)

(b) Show that $(2, 0)$ is a stationary point. (1 mark)

(c) Evaluate $\int_0^2 \left(x^3 - 3x^2 + 4\right) dx$. (2 marks)

130. Let $f(x) = x^2e^x$.

(a) Show that $f'(x) = xe^x(x + 2)$. (2 marks)

(b) Find the exact coordinates of the stationary points. (2 marks)

(c) Determine the nature of the stationary point at $x = 0$. (1 mark)

131. Solve the following equations.

(a) $\log_2(x + 3) = 4$ (2 marks)

(b) $3^{2x} = 81$ (1 mark)

(c) $\log_5 x + \log_5(x - 4) = 1$ (2 marks)

6 Integrating to logarithms

ANCHOR QUESTION

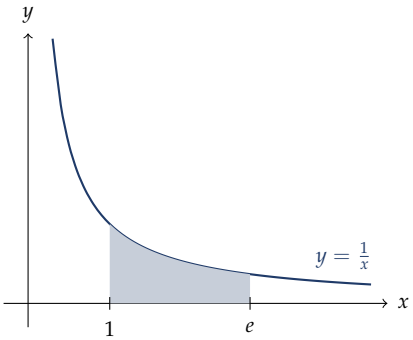
Find $\int \left(\dfrac{10}{2x+3} - \dfrac{4}{x} \right) dx.$

SYLLABUS MAV-12-05

- Derive and use the formula $\int \dfrac{1}{x}\, dx = \ln|x| + C$ where $x \neq 0$
- Establish and use the formula $\int \dfrac{1}{ax+b}\, dx = \dfrac{1}{a} \ln|ax+b| + C$, where a and b are constants and $a \neq 0$
- Establish and use $\int \dfrac{u'}{u}\, dx = \ln|u| + C$, where u is a function of x, or $\int \dfrac{f'(x)}{f(x)} dx = \ln|f(x)| + C$, on a domain where $f(x) \neq 0$

NOTES

- $\int \dfrac{1}{x}\, dx = \ln|x| + C$. It fills the gap in the power rule, which fails at x^{-1}.
- $\int \dfrac{1}{ax+b}\, dx = \dfrac{1}{a} \ln|ax+b| + C$: divide by the coefficient of x.
- $\int \dfrac{f'(x)}{f(x)}\, dx = \ln|f(x)| + C$: the top is the derivative of the bottom. This is on the reference sheet.
- If the top is $k f'(x)$, write the constant k in front: $\int \dfrac{x}{x^2+4}\, dx = \frac{1}{2} \ln\left(x^2+4\right) + C$.
- Use $\ln a - \ln b = \ln \dfrac{a}{b}$ and $\ln 1 = 0$ to tidy answers to definite integrals.
- If the bottom has no matching derivative on top, divide the terms out first where possible: $\dfrac{x^2+3x+1}{x} = x + 3 + \dfrac{1}{x}$.



WORKED EXAMPLE

Find $\int \left(\dfrac{12}{3x-2} + \dfrac{5}{x} \right) dx.$

$\int \dfrac{12}{3x-2} dx = \dfrac{12}{3} \ln|3x-2| = 4\ln|3x-2|$ divide by $a = 3$

$\int \dfrac{5}{x}\, dx = 5\ln|x|$

Total: $4\ln|3x-2| + 5\ln|x| + C$

Tier 1 · $\frac{1}{x}$

132. Find each integral.

(a) $\displaystyle\int \frac{7}{x}\,dx$

(b) $\displaystyle\int \frac{1}{2x}\,dx$

(c) $\displaystyle\int \left(\frac{1}{x} + x\right) dx$

133. Show by differentiation that $\ln x$ is a primitive of $\frac{1}{x}$ for $x > 0$.

134. Evaluate $\displaystyle\int_1^4 \frac{3}{x}\,dx$.

Tier 2 · $\frac{1}{ax+b}$

135. Find each integral.

(a) $\displaystyle\int \frac{1}{3x+1}\,dx$

(b) $\displaystyle\int \frac{6}{3x-1}\,dx$

(c) $\displaystyle\int \frac{1}{1-x}\,dx$

136. Find $\displaystyle\int \left(\frac{2}{4x-3} + e^{2x}\right) dx$.

137. Evaluate $\displaystyle\int_0^2 \frac{1}{2x+1}\,dx$.

Tier 3 · $\frac{f'(x)}{f(x)}$

138. For each fraction, write $f(x)$, $f'(x)$ and k so that the integrand is $\dfrac{k\,f'(x)}{f(x)}$.

(a) $\dfrac{x}{x^2+4}$

(b) $\dfrac{x^2}{x^3+1}$

(c) $\dfrac{2x+3}{x^2+3x}$

139. Find $\displaystyle\int \frac{2x}{x^2+1}\,dx$ and $\displaystyle\int \frac{x}{x^2-9}\,dx$.

140. Find $\displaystyle\int \frac{x^2}{2x^3+5}\,dx$.

Tier 4 · definite integrals of $\dfrac{f'}{f}$

141. Find $\displaystyle\int \frac{x}{4+x^2}\,dx$.

142. Evaluate $\displaystyle\int_0^1 \frac{6x^2}{x^3+1}\,dx$ in exact form.

143. Show that $\displaystyle\int_2^5 \frac{x}{x^2-1}\,dx = \tfrac{1}{2}\ln 8$.

Tier 5 · split the fraction first

144. Find $\displaystyle\int \frac{x^2+3x+1}{x}\,dx$.

145. Find the exact value of $\displaystyle\int_1^e \left(\frac{1}{x}+\frac{2}{x^2}\right)dx$.

146. Find $\displaystyle\int \frac{6}{2x+1} - \frac{3}{x} + 4x\,dx$.

Conceptual

147. Prove that $\displaystyle\int \frac{f'(x)}{f(x)}\,dx = \ln|f(x)| + C$.

148. Explain why the answer to $\displaystyle\int \frac{1}{x}\,dx$ cannot be found using $\dfrac{x^{n+1}}{n+1}$.

149. A student writes $\displaystyle\int \frac{1}{x^2+1}\,dx = \ln\left(x^2+1\right) + C$. Explain the error.

150. It is given that $\dfrac{f'(x)}{f(x)} = 2x$, where $f(0) = 1$ and $f(x) > 0$ for all x.

(a) Show that $\ln f(x) = x^2 + C$. (1 mark)

(b) Find $f(x)$. (2 marks)

151. Show that $\displaystyle\int_0^{\pi/2} \frac{\cos x}{1+\sin x}\,dx = \ln 2$.

Revision

152. Let $f(x) = x^2e^{-x}$.

(a) Show that $f'(x) = xe^{-x}(2-x)$. (2 marks)

(b) Find the stationary points. (2 marks)

(c) Find $\displaystyle\int_0^1 \left(e^x + e^{-x}\right) dx$. (1 mark)

153. A curve has $\dfrac{dy}{dx} = \dfrac{4}{2x+1}$ and passes through $(0,3)$.

(a) Find the equation of the curve, for $x > -\frac{1}{2}$. (2 marks)

(b) Find the exact y-coordinate when $x = 2$. (1 mark)

(c) Show that the curve is always increasing. (1 mark)

154. The first term of a geometric series is 5 and the fourth term is -40.

(a) Show that the common ratio is -2. (2 marks)

(b) Find the sum of the first 10 terms. (2 marks)

(c) Explain why the series has no limiting sum. (1 mark)

7 Integrating trigonometric functions

ANCHOR QUESTION

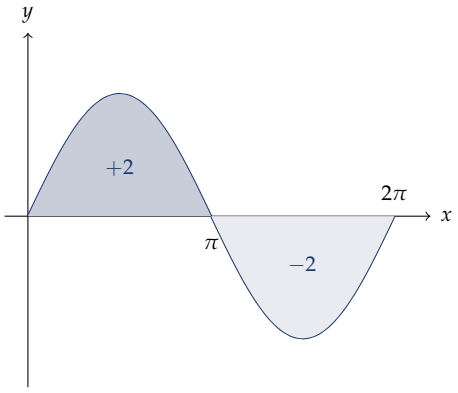
Find $\int (12\cos 3x - 4\sin 2x)\,dx$.

SYLLABUS MAV-12-05

- Establish and use the formulas $\int \sin x\,dx = -\cos x + C$, $\int \cos x\,dx = \sin x + C$ and $\int \sec^2 x\,dx = \tan x + C$
- Establish and use indefinite integrals of the form $\int f(ax+b)\,dx$, where a and b are constants and $a \neq 0$, and $f(x) = \sin x$, $f(x) = \cos x$ and $f(x) = \sec^2 x$
- Determine indefinite integrals of the form $\int f'(x)\sin f(x)\,dx$, $\int f'(x)\cos f(x)\,dx$ and $\int f'(x)\sec^2 f(x)\,dx$

NOTES

- $\int \cos x\,dx = \sin x + C$, $\int \sin x\,dx = -\cos x + C$, $\int \sec^2 x\,dx = \tan x + C$. Angles are in radians.
- Check each by differentiating: $\dfrac{d}{dx}(-\cos x) = \sin x$, so the minus sign belongs to $\sin x$.
- With $ax+b$: divide by a and keep the sign. $\int \cos(ax+b)\,dx = \dfrac{1}{a}\sin(ax+b) + C$, $\int \sin(ax+b)\,dx = -\dfrac{1}{a}\cos(ax+b) + C$, $\int \sec^2(ax+b)\,dx = \dfrac{1}{a}\tan(ax+b) + C$.
- $\int f'(x)\cos f(x)\,dx = \sin f(x) + C$, $\int f'(x)\sin f(x)\,dx = -\cos f(x) + C$, $\int f'(x)\sec^2 f(x)\,dx = \tan f(x) + C$.
- Exact values for evaluating: $\sin\frac{\pi}{6} = \frac{1}{2}$, $\cos\frac{\pi}{3} = \frac{1}{2}$, $\sin\frac{\pi}{4} = \cos\frac{\pi}{4} = \frac{\sqrt{2}}{2}$, $\tan\frac{\pi}{4} = 1$, $\tan\frac{\pi}{3} = \sqrt{3}$.



WORKED EXAMPLE

Find $\int (10\sin 5x + 6\sec^2 2x)\,dx$.

$\int 10\sin 5x\,dx = -\dfrac{10}{5}\cos 5x = -2\cos 5x$ divide by $a = 5$, keep the minus

$\int 6\sec^2 2x\,dx = \dfrac{6}{2}\tan 2x = 3\tan 2x$

Total: $-2\cos 5x + 3\tan 2x + C$

Tier 1 · the three basic integrals

155. Find each integral.

(a) $\displaystyle\int \cos x\,dx$

(b) $\displaystyle\int \sin x\,dx$

(c) $\displaystyle\int \sec^2 x\,dx$

156. Find $\displaystyle\int (3\cos x - 2\sin x)\,dx$.

157. Show by differentiation that $-\cos x$ is a primitive of $\sin x$.

Tier 2 · $ax + b$

158. Find each integral.

(a) $\displaystyle\int \cos 4x\,dx$

(b) $\displaystyle\int \sin 2x\,dx$

(c) $\displaystyle\int \sec^2 3x\,dx$

159. Find each integral.

(a) $\displaystyle\int 6\sin(3x+1)\,dx$

(b) $\displaystyle\int \sin\frac{x}{2}\,dx$

(c) $\displaystyle\int \cos\left(2x - \frac{\pi}{3}\right)dx$

160. Find $\displaystyle\int \left(8\cos 2x - 3\sin 4x + \sec^2 x\right)dx$.

Tier 3 · definite integrals

161. Evaluate each.

(a) $\displaystyle\int_0^{\pi/2} \cos x\,dx$

(b) $\displaystyle\int_0^{\pi} \sin x\,dx$

(c) $\displaystyle\int_0^{\pi/4} \sec^2 x\,dx$

162. Evaluate $\displaystyle\int_0^{\pi/4} \cos 2x\,dx$ and $\displaystyle\int_0^{\pi/2} \sin 2x\,dx$.

163. Evaluate $\displaystyle\int_0^{\pi/4} \sec^2 x\,dx$.

Tier 4 · combined and shifted

164. Evaluate $\displaystyle\int_0^{\pi/4} \left(\cos 2x + \sec^2 x\right) dx$.

165. Evaluate $\displaystyle\int_0^{\pi/6} \left(3\cos 3x + \sec^2 2x\right) dx$ in exact form.

166. Find the exact value of $\displaystyle\int_{\pi/6}^{\pi/2} \sin\left(x - \frac{\pi}{6}\right) dx$.

Tier 5 · $f'(x)$ with a trig function of f

167. Find each integral.

(a) $\displaystyle\int x\cos\left(x^2\right) dx$

(b) $\displaystyle\int 3x^2 \sin\left(x^3\right) dx$

(c) $\displaystyle\int \sec^2 x\, e^{\tan x} dx$

168. Find each integral.

(a) $\displaystyle\int \cos x\, \sin^4 x\, dx$

(b) $\displaystyle\int \sec^2 x\, \tan^3 x\, dx$

(c) $\displaystyle\int \sin x\, \cos^2 x\, dx$

169. Evaluate $\displaystyle\int_0^{\pi/2} \sin x \cos^2 x\, dx$.

Tier 6 · trigonometric quotients

170. Find $\displaystyle\int \frac{\cos x}{\sin x} dx$ and $\displaystyle\int \tan x\, dx$.

171. Evaluate $\displaystyle\int_{\pi/6}^{\pi/2} \frac{\cos x}{\sin x} dx$.

172. Show that $\displaystyle\int_0^{\pi/3} \tan x\, dx = \ln 2$.

Conceptual

173. Prove that $\int \sin(ax+b)\,dx = -\frac{1}{a}\cos(ax+b) + C$ for $a \neq 0$.

174. Explain why $\int_0^{2\pi} \sin x\,dx = 0$ even though the graph is not on the axis.

175. A student writes $\int \cos 3x\,dx = 3\sin 3x + C$. Explain the error.

176. Given $\frac{dy}{dx} = -6\sin 2x$ and $y(0) = 5$.

(a) Find y. (2 marks)

(b) Find the greatest value of y and the smallest positive x where it occurs. (2 marks)

177. What is $\int \sin\left(2x + \frac{\pi}{3}\right)\,dx$?

A. $-2\cos\left(2x+\frac{\pi}{3}\right) + C$ B. $-\frac{1}{2}\cos\left(2x+\frac{\pi}{3}\right) + C$ C. $\frac{1}{2}\cos\left(2x+\frac{\pi}{3}\right) + C$ D. $2\cos\left(2x+\frac{\pi}{3}\right) + C$

Revision

178. Let $y = x\sin x$.

(a) Find $\frac{dy}{dx}$. (2 marks)

(b) Find the equation of the tangent at $x = \frac{\pi}{2}$. (2 marks)

(c) Show that $\int_0^{\pi} \cos x\,dx = 0$. (1 mark)

179. Show that $\frac{1}{\cos\theta} - \cos\theta = \sin\theta\tan\theta$.

(a) Prove the identity. (2 marks)

(b) Hence solve $\frac{1}{\cos\theta} - \cos\theta = \sin\theta$ for $0 \leq \theta \leq 2\pi$. (3 marks)

180. Let $f(x) = x\left(x^2+1\right)^3$.

(a) Find $\int f(x)\,dx$. (2 marks)

(b) Find $\int \frac{2x}{x^2+1}\,dx$. (1 mark)

(c) A curve has $\frac{dy}{dx} = \frac{2x}{x^2+1}$ and passes through $(0,2)$. Find y when $x = 1$. (2 marks)

8 Differentiate, then integrate

ANCHOR QUESTION

Differentiate xe^{4x}. Hence find $\int xe^{4x}\,dx$.

SYLLABUS MAV-12-05

- Find primitives of functions involving exponential, logarithmic and trigonometric functions
- Use algebraic manipulation to express given functions in forms suitable for determining primitive functions
- Apply $\int_a^b f(x)\,dx = F(b) - F(a)$, where $F(x)$ is a primitive of $f(x)$, to calculate definite integrals and solve related theoretical problems involving functions within the scope of the Mathematics Advanced course

NOTES

- If $\frac{d}{dx}F(x) = g(x)$, then $\int g(x)\,dx = F(x) + C$. A derivative you have found is a free integral.
- 'Hence' means use the result you have just found.
- Differentiate the given function (product, quotient or chain rule). The derivative has two parts: the integral you want and an easy one.
- Integrate both sides, move the easy integral across, then divide to leave the wanted integral on its own.
- For a definite integral, write the primitive in brackets and substitute the limits. The $+C$ is not needed.
- If the question says 'show that', work from the left side to the right side and state the result.

$$\frac{d}{dx}F(x) = f(x) \xrightarrow{\text{hence}} \int f(x)\,dx = F(x) + C$$

WORKED EXAMPLE

Differentiate xe^{5x}. Hence find $\int xe^{5x}\,dx$.

$$\frac{d}{dx}\left(xe^{5x}\right) = e^{5x} + 5xe^{5x}$$ product rule

Integrate both sides: $xe^{5x} = \frac{1}{5}e^{5x} + 5\int xe^{5x}dx$

$$\int xe^{5x}dx = \tfrac{1}{5}\left(xe^{5x} - \tfrac{1}{5}e^{5x}\right) + C = \frac{xe^{5x}}{5} - \frac{e^{5x}}{25} + C$$

Tier 1 · integrate a derivative

181. Given $\dfrac{d}{dx}\left(xe^{3x}\right) = e^{3x} + 3xe^{3x}$, find $\displaystyle\int \left(e^{3x} + 3xe^{3x}\right) dx$.

182. Given $\dfrac{d}{dx}\left(x^2 \sin x\right) = 2x\sin x + x^2\cos x$, find $\displaystyle\int \left(2x\sin x + x^2\cos x\right) dx$.

183. Given $\dfrac{d}{dx}\left(x^2+1\right)^3 = 6x\left(x^2+1\right)^2$, find $\displaystyle\int x\left(x^2+1\right)^2 dx$.

Tier 2 · the wanted integral and the easy one

184. From $\displaystyle\int \left(e^{3x} + 3xe^{3x}\right) dx = xe^{3x} + C$, find $\displaystyle\int xe^{3x} dx$.

185. From $\displaystyle\int (\sin x + x\cos x)\, dx = x\sin x + C$, find $\displaystyle\int x\cos x\, dx$.

186. From $\displaystyle\int (\ln x + 1)\, dx = x\ln x + C$, find $\displaystyle\int \ln x\, dx$.

Tier 3 · differentiate, then integrate

187. Differentiate $x\cos x$. Hence find $\displaystyle\int x\sin x\, dx$.

188. Differentiate $x\sin 2x$. Hence find $\displaystyle\int x\cos 2x\, dx$.

189. Differentiate xe^{-2x}. Hence find $\displaystyle\int xe^{-2x} dx$.

Tier 4 · logarithms

190. Differentiate $x\ln x$. Hence find $\displaystyle\int \ln x\, dx$.

191. Show that $\dfrac{d}{dx}\left(x^2\ln x\right) = 2x\ln x + x$. Hence find $\displaystyle\int x\ln x\, dx$.

192. Differentiate $x^3\ln x$. Hence find $\displaystyle\int x^2\ln x\, dx$.

Tier 5 · powers and quotients

193. Differentiate $(x^2+3)^5$. Hence find $\int x\,(x^2+3)^4\,dx$.

194. Differentiate $e^{3x}(3x-1)$. Hence find $\int xe^{3x}dx$.

195. Differentiate $\dfrac{x}{x^2+1}$. Hence find $\displaystyle\int \frac{1-x^2}{(x^2+1)^2}\,dx$.

Tier 6 · definite integrals and 'show that'

196. Show that $\dfrac{d}{dx}(\sin x - x\cos x) = x\sin x$. Hence evaluate $\displaystyle\int_0^{\pi} x\sin x\,dx$.

197. Show that $\dfrac{d}{dx}(\tan x - x) = \tan^2 x$. Hence evaluate $\displaystyle\int_0^{\pi/4} \tan^2 x\,dx$.

198. Differentiate $e^{-x}(x+1)$. Hence find the exact value of $\displaystyle\int_0^1 xe^{-x}dx$.

Conceptual

199. A student answers 'Differentiate xe^{2x}. Hence find $\int xe^{2x}dx$' with $\int xe^{2x}dx = xe^{2x}+C$. Explain the error.

200. Explain what the word 'hence' asks you to do.

201. Show that $\dfrac{d}{dx}(x\ln x - x) = \ln x$. What does this tell you about $\displaystyle\int_1^e \ln x\,dx$?

202. Consider $y = x^2\ln x$ for $x>0$.

(a) Show that $\dfrac{dy}{dx} = 2x\ln x + x$. (2 marks)

(b) Hence find $\displaystyle\int_1^e x\ln x\,dx$. (3 marks)

203. Show that $\dfrac{d}{dx}\left(x\sqrt{x+1}\right) = \dfrac{3x+2}{2\sqrt{x+1}}$.

Revision

204. Let $f(x) = x^3 - 6x^2 + 9x$.

(a) Find the x-coordinates of the stationary points. (2 marks)

(b) Show that $f(x) \geq 0$ for $0 \leq x \leq 3$. (1 mark)

(c) Find the area between the curve and the x-axis from $x = 0$ to $x = 3$. (2 marks)

205. Events A and B are independent, with $P(A) = 0.4$ and $P(B) = 0.5$.

(a) Find $P(A \cap B)$. (1 mark)

(b) Find $P(A \cup B)$. (1 mark)

(c) Find $P(A \mid B)$ and explain the result. (2 marks)

206. Let $f(x) = \dfrac{3}{2x - 1} + e^{-x}$.

(a) Find $\displaystyle\int f(x)\,dx$ for $x > \tfrac{1}{2}$. (2 marks)

(b) Find the exact value of $\displaystyle\int_1^2 f(x)\,dx$. (2 marks)

9 Mixed integration: choosing the integral

ANCHOR QUESTION

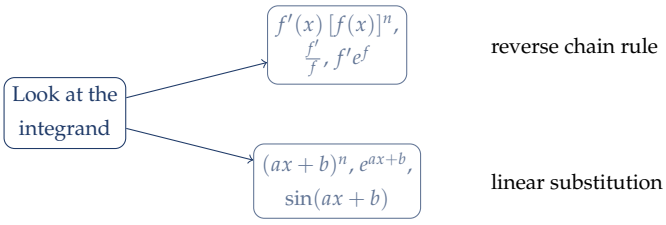
Find $\int \left(\dfrac{9x^2}{x^3+2} - 4e^{2x} + 3\sin x \right) dx.$

SYLLABUS MAV-12-05

- Use the identities $\int (f(x)+g(x))dx = \int f(x)\,dx + \int g(x)\,dx$ and $\int kf(x)\,dx = k\int f(x)\,dx$
- Use algebraic manipulation to express given functions in forms suitable for determining primitive functions
- Find primitives of functions involving exponential, logarithmic and trigonometric functions, and of functions of the form $f'(x)\left[f(x)\right]^n$
- Determine $f(x)$, given $f'(x)$ and an initial condition $f(a) = b$

NOTES

- Look at the form first. x^n or $(ax+b)^n$: add 1 to the power. e^{ax+b}, $\sin(ax+b)$, $\cos(ax+b)$, $\sec^2(ax+b)$: divide by a.
- $\dfrac{1}{ax+b}$: $\dfrac{1}{a}\ln|ax+b|$. Top is the derivative of the bottom: $\ln|f(x)|$.
- $f'e^f$: e^f. $f'\cos f$: $\sin f$. $f'\sin f$: $-\cos f$. $f'f^n$: $\dfrac{f^{n+1}}{n+1}$.
- If no form fits: expand brackets, split a fraction into terms, or rewrite with index laws, then look again.
- Integrate each term separately and check the answer by differentiating.
- A definite integral or an initial condition needs the answer in full first: find the primitive, then substitute.



WORKED EXAMPLE

Find $\int \left(\dfrac{6x}{3x^2+2} + 8e^{-4x} + \dfrac{2}{x^2} \right) dx.$

$\dfrac{6x}{3x^2+2}$ is $\dfrac{f'}{f}$ with $f = 3x^2+2$: $\ln\left(3x^2+2\right)$.

$8e^{-4x}$ is e^{ax+b}: $-2e^{-4x}$. $\quad 2x^{-2}$ is a power: $-\dfrac{2}{x}$.

Total: $\ln\left(3x^2+2\right) - 2e^{-4x} - \dfrac{2}{x} + C$

Tier 1 · name the form

207. Name the form of each integrand: power, e^{ax+b}, $\dfrac{1}{ax+b}$, $\dfrac{f'}{f}$, $f'e^{f}$ or $f'f^{n}$.

(a) $\dfrac{3}{5x-2}$

(b) e^{-4x}

(c) $\dfrac{2x}{x^2+7}$

(d) $x^2e^{x^3}$

(e) $3x\left(x^2+1\right)^4$

208. Name the form of each integrand.

(a) $\sin 5x$

(b) $\dfrac{x}{x^2-4}$

(c) $\dfrac{1}{x^3}$

(d) $(4x-1)^5$

(e) $\sec^2\dfrac{x}{2}$

209. Which integrand does *not* need any rewriting? $\dfrac{x+1}{x}$, $\dfrac{1}{e^x}$, e^{3x}, $(x+1)^2$.

Tier 2 · sums of different forms

210. Find $\displaystyle\int\left(\frac{2}{3x+1}-\sin 2x\right)dx$.

211. Find $\displaystyle\int\left(\sec^2 3x+\frac{1}{\sqrt{x}}\right)dx$.

212. Find $\displaystyle\int\left(\cos\frac{x}{2}+e^{\frac{x}{2}}-3x^2\right)dx$.

Tier 3 · algebra first

213. Find $\displaystyle\int\frac{x^2+3x+1}{x}\,dx$.

214. Find $\displaystyle\int\frac{e^{2x}+1}{e^x}\,dx$.

215. Find $\displaystyle\int\frac{\left(x^2-1\right)^2}{x^2}\,dx$.

216. Find $\displaystyle\int\left(e^x+1\right)^2dx$.

Tier 4 · recognising $f'(x)$

217. Find $\displaystyle\int \left(x^2 e^{x^3} + \frac{x^2}{x^3+1}\right) dx$.

218. Find $\displaystyle\int x\,(x^2-4)^3\,dx$ and $\displaystyle\int \frac{(\ln x)^2}{x}\,dx$.

219. Find $\displaystyle\int \frac{e^x}{e^x+1}\,dx$ and $\displaystyle\int \frac{\sin x}{2+\cos x}\,dx$.

Tier 5 · definite integrals

220. Evaluate $\displaystyle\int_0^{\pi/4} \left(\cos 2x + \sec^2 x\right) dx$.

221. Evaluate $\displaystyle\int_1^2 \left(\frac{4x^3}{x^4+1} + e^{2-x}\right) dx$ in exact form.

222. Evaluate $\displaystyle\int_1^e \left(\frac{1}{x} + x\right) dx$ and $\displaystyle\int_0^1 \left(e^{3x} - e^{-3x}\right) dx$.

Tier 6 · finding the function

223. A curve has $\dfrac{dy}{dx} = \dfrac{6x}{x^2+1} + e^{-2x}$ and passes through $(0,2)$. Find its equation.

224. Given $f'(x) = \cos 2x + \dfrac{1}{(x+1)^2}$ and $f(0) = 1$, find $f(x)$.

225. Given $f'(x) = 2^x$ and $f(0) = 3$, find $f(x)$.

Conceptual

226. Explain why $\displaystyle\int e^{x^2} dx$ cannot be found by the methods in this booklet.

227. A student writes $\displaystyle\int \frac{1}{x^2+1}\,dx = \ln\left(x^2+1\right) + C$. Explain the error.

228. A student writes $\displaystyle\int \frac{x+1}{x}\,dx = \frac{\frac{1}{2}x^2 + x}{\ln x} + C$. Explain the error and correct it.

229. Show that $\displaystyle\int_0^1 \left(\frac{2x}{x^2+1} + \frac{1}{2x+1}\right) dx = \ln 2 + \tfrac{1}{2}\ln 3$.

230. What is $\displaystyle\int \frac{3x^2}{x^3+5}\,dx$?

A. $\ln\left|x^3+5\right| + C$ B. $\dfrac{x^3}{x^3+5} + C$ C. $3x^2 \ln\left|x^3+5\right| + C$ D. $\dfrac{1}{x^3+5} + C$

Revision

231. Let $f(x) = \dfrac{\ln x}{x}$ for $x > 0$.

(a) Show that $f'(x) = \dfrac{1 - \ln x}{x^2}$. (2 marks)

(b) Find the stationary point. (2 marks)

(c) Hence evaluate $\displaystyle\int_1^e \frac{1 - \ln x}{x^2}\,dx$. (1 mark)

232. Evaluate $\displaystyle\int_{-\pi/3}^{\pi/3} (\sin x + \cos x)\,dx$.

(a) State whether $\sin x$ and $\cos x$ are odd or even. (1 mark)

(b) Hence evaluate the integral. (2 marks)

233. Let $f(x) = x^2 - 4x + 3$.

(a) Find the roots of $f(x) = 0$. (1 mark)

(b) Find the coordinates of the vertex. (1 mark)

(c) Solve $f(x) = 3$. (1 mark)

(d) Find the range of $f(x)$. (1 mark)

10 Areas between a curve and the x-axis

ANCHOR QUESTION

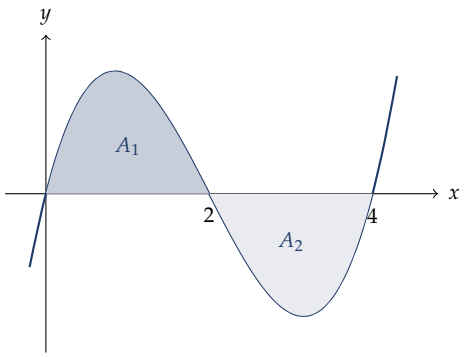
Find the total area between $y = x(x-1)(x-3)$ and the x-axis from $x = 0$ to $x = 3$.

SYLLABUS MAV-12-05

- Describe, where $f(x) \geq 0$ on $a \leq x \leq b$, the area bounded by the graph of $y = f(x)$, the x-axis and the lines $x = a$ and $x = b$ as $\int_a^b f(x)\,dx$
- Recognise, where $f(x) \leq 0$ on $a \leq x \leq b$, the area as $\left|\int_a^b f(x)\,dx\right|$ or $-\int_a^b f(x)\,dx$
- Conclude that $\int_a^b f(x)\,dx =$ (area of regions above the x-axis) $-$ (area of regions below the x-axis)
- Use definite integrals to solve problems involving areas of regions bounded by $y = f(x)$, the x-axis and the lines $x = a$ and $x = b$, where $f(x) \geq 0$, $f(x) \leq 0$, or $f(x)$ changes sign, with or without the graph provided

NOTES

- Above the axis ($f \geq 0$): area $= \int_a^b f(x)\,dx$. Below the axis ($f \leq 0$): area $= -\int_a^b f(x)\,dx$.
- The integral is signed: $\int_a^b f(x)\,dx =$ (area above) $-$ (area below). It can be 0 when the area is not.
- If the curve crosses the axis between a and b: find the x-intercepts, integrate each section separately, make each positive, then add.
- Find intercepts by solving $f(x) = 0$ and keep only those with $a < x < b$.
- Symmetry helps: for an odd f, the two halves of $\int_{-a}^{a}$ cancel, but their areas add.
- Sketch first if no graph is given: the sketch shows which sections are above or below.



WORKED EXAMPLE

Find the area enclosed by $y = x^2 - 9$ and the x-axis.

The curve meets the axis at $x = \pm 3$ and lies below it between them. solve $x^2 - 9 = 0$

$$\int_{-3}^{3} (x^2 - 9)\,dx = \left[\frac{x^3}{3} - 9x\right]_{-3}^{3} = (9 - 27) - (-9 + 27) = -36$$

Area $= 36$ square units

Tier 1 · integral or area?

234. For each, state whether the integral is positive, negative or zero, and give the area between the graph and the x-axis.

(a) $\displaystyle\int_0^2 x\,dx = 2$

(b) $\displaystyle\int_{-2}^0 x\,dx = -2$

(c) $\displaystyle\int_{-2}^2 x\,dx = 0$

235. For each, state the sign of the integral and the area.

(a) $\displaystyle\int_1^3 2\,dx = 4$

(b) $\displaystyle\int_0^3 (-1)\,dx = -3$

(c) $\displaystyle\int_{-1}^1 x^3 dx = 0$

236. True or false: the area between a curve and the x-axis is always $\left|\displaystyle\int_a^b f(x)\,dx\right|$. Explain.

Tier 2 · area above the axis

237. Find the area between the curve, the x-axis and the given lines.

(a) $y = 3x^2$, $x = 1$, $x = 2$

(b) $y = e^x$, $x = 0$, $x = \ln 3$

(c) $y = \dfrac{1}{x}$, $x = 1$, $x = 4$

238. Find the area between the curve, the x-axis and the given lines.

(a) $y = \cos x$, $x = 0$, $x = \dfrac{\pi}{2}$

(b) $y = e^{2x}$, $x = 0$, $x = 1$

(c) $y = \dfrac{4}{2x+1}$, $x = 0$, $x = 2$

239. Find the area enclosed by $y = 9 - x^2$ and the x-axis.

Tier 3 · area below the axis

240. Find the area between $y = -x^2$, the x-axis and the lines $x = 0$ and $x = 3$.

241. Find the area enclosed by $y = x^2 - 6x$ and the x-axis.

242. Find the area between $y = -\dfrac{3}{x}$, the x-axis and $x = 1$, $x = e$.

Tier 4 · where does the curve cross the axis?

243. Find the x-intercepts that lie inside the given interval.

(a) $y = x(x-3)$ on $[0, 4]$

(b) $y = x^2 - 1$ on $[0, 3]$

(c) $y = \sin x$ on $[0, 2\pi]$

244. Find the x-intercept inside each interval.

(a) $y = x^2 - 4$ on $[0, 5]$

(b) $y = 2x - 6$ on $[0, 5]$

(c) $y = \cos x$ on $[0, \pi]$

(d) $y = e^x - 2$ on $[0, 2]$

245. Without calculating, decide whether the total area for $y = x^3 - x$ on $[-1, 1]$ equals $\displaystyle\int_{-1}^{1} (x^3 - x)\,dx$. Explain.

Tier 5 · split and add

246. Find the total area between $y = x(x-2)(x-4)$ and the x-axis from $x = 0$ to $x = 4$.

247. Find the total area between $y = x(x-3)$ and the x-axis from $x = 0$ to $x = 5$.

248. Find the total area between $y = e^x - 2$, the x-axis and the lines $x = 0$ and $x = \ln 4$.

249. Find the total area between $y = x^3 - x$ and the x-axis from $x = -1$ to $x = 1$.

Tier 6 · composite regions and 'show that'

250. The region lies under $y = 2x$ for $0 \le x \le 1$ and under $y = \dfrac{2}{x}$ for $1 \le x \le 3$. Find its exact area.

251. The area between $y = \dfrac{4}{2x+1}$, the x-axis, $x = 0$ and $x = k$ is $2\ln 5$. Find $k > 0$.

252. Show that the area enclosed by $y = 9 - 3^x$ and the coordinate axes is $18 - \dfrac{8}{\ln 3}$.

Conceptual

253. Explain why $\int_0^{2\pi} \sin x\,dx = 0$ but the area between $y = \sin x$ and the x-axis over $[0, 2\pi]$ is 4.

254. Use symmetry to find the total area between $y = x^3$ and the x-axis from $x = -2$ to $x = 2$.

255. A student finds the area between $y = x^2 - 4$ and the x-axis from $x = 0$ to $x = 3$ as $\int_0^3 (x^2 - 4)\,dx = -3$ and writes 3. Explain the error and find the correct area.

256. Let $f(x) = x^2 - 4x$.

(a) Find the x-intercepts. (1 mark)

(b) Find $\int_1^4 f(x)\,dx$ and $\int_4^5 f(x)\,dx$. (2 marks)

(c) Find the total area between the curve and the x-axis from $x = 1$ to $x = 5$. (1 mark)

257. Which gives the area between $y = f(x)$ and the x-axis for $0 \le x \le 3$, if $f(x) \ge 0$ on $[0, 2]$ and $f(x) \le 0$ on $[2, 3]$?

A. $\int_0^3 f$ B. $\int_0^2 f - \int_2^3 f$ C. $\int_0^2 f + \int_2^3 f$ D. $\left|\int_0^3 f\right|$

Revision

258. Let $f(x) = e^{2x} - 4$.

(a) Find the x-intercept. (1 mark)

(b) Find $\int_0^{\ln 2} f(x)\,dx$. (2 marks)

(c) Hence find the area between the curve, the axes and $x = \ln 2$. (1 mark)

259. In triangle ABC, $a = 7$, $b = 5$ and $C = 60°$.

(a) Find the exact length of c. (2 marks)

(b) Find the exact area of the triangle. (2 marks)

260. Consider $\int \cos x \sin^2 x\,dx$.

(a) Find the integral. (1 mark)

(b) Evaluate $\int_0^{\pi/2} \cos x \sin^2 x\,dx$. (2 marks)

(c) Explain why this is also the area between $y = \cos x \sin^2 x$ and the x-axis from 0 to $\frac{\pi}{2}$. (1 mark)

11 Areas to the y-axis

ANCHOR QUESTION

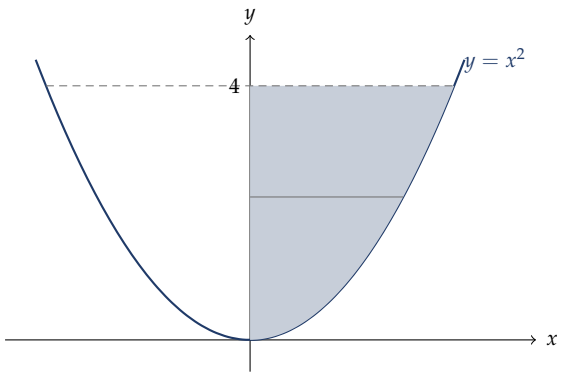
Find the area bounded by $y = x^3$ (for $x \geq 0$), the y-axis and the line $y = 8$.

SYLLABUS MAV-12-05

- Use definite integrals to solve problems involving the areas of regions bounded by the graph of the continuous function $y = f(x)$, the y-axis and the lines $y = a$ and $y = b$, in cases where $x \geq 0$ throughout $a \leq y \leq b$, $x \leq 0$ throughout $a \leq y \leq b$ or where x changes sign in the interval, with or without the graph provided
- Use the fact that the graphs of $y = a^x$ and $y = \log_a x$ are reflections of each other in the line $y = x$ to solve problems involving areas between the x-axis or y-axis and a curve involving either an exponential or logarithmic function

NOTES

- For a region between a curve and the y-axis, use horizontal strips: $A = \int_c^d x\,dy$, with limits that are y-values.
- Write the curve with x as the subject: $y = x^2$ $(x \geq 0)$ gives $x = \sqrt{y}$; $y = \ln x$ gives $x = e^y$; $y = e^x$ gives $x = \ln y$.
- If $x \geq 0$ the area is $\int_c^d x\,dy$. If $x \leq 0$ it is $-\int_c^d x\,dy$. If x changes sign, split at the y-value where $x = 0$.
- Alternative: the rectangle containing the region, minus the area on the other side of the curve (found with dx).
- $y = a^x$ and $y = \log_a x$ are reflections in $y = x$: an area between $y = \log_a x$ and the x-axis is an area between $y = a^x$ and the y-axis.



WORKED EXAMPLE

Find the exact area between $y = \ln x$, the y-axis and the lines $y = 0$ and $y = 2$.

Make x the subject: $x = e^y$. the limits are y-values

$A = \int_0^2 e^y\,dy = \left[e^y\right]_0^2 = e^2 - 1$ square units

Tier 1 · x as a function of y (take $x \geq 0$)

261. Write x in terms of y.

(a) $y = x^2$

(b) $y = x^3$

(c) $y = \dfrac{1}{x}$

(d) $y = x^2 + 1$

262. Write x in terms of y.

(a) $y = x^2 + 4$

(b) $y = 3x$

(c) $y = \ln x$

(d) $y = e^x$

(e) $y = \sqrt{x}$

263. For $y = 2x + 1$, write the integral $\displaystyle\int x\,dy$ for the area between the line, the y-axis, $y = 1$ and $y = 5$, then evaluate it.

Tier 2 · areas with respect to y

264. Find the area between the curve, the y-axis and the given lines.

(a) $y = x^2$ $(x \geq 0)$, $y = 1$, $y = 4$

(b) $y = x^3$ $(x \geq 0)$, $y = 1$, $y = 8$

(c) $y = \sqrt{x}$, $y = 1$, $y = 3$

265. Find the area bounded by $y = x^3$ (for $x \geq 0$), the y-axis and the line $y = 8$.

266. Find the area between $y = 2x$, the y-axis and the lines $y = 0$ and $y = 6$.

Tier 3 · shifted curves

267. Find the exact area between $y = x^2 + 1$ $(x \geq 0)$, the y-axis and the lines $y = 2$ and $y = 5$.

268. Find the exact area between $y = x^2 + 3$ $(x \geq 0)$, the y-axis and the lines $y = 4$ and $y = 7$.

269. Find the exact area between $y = \dfrac{2}{x}$, the y-axis and the lines $y = 1$ and $y = 4$.

Tier 4 · exponential and logarithmic curves

270. Find the exact area between $y = \ln x$, the y-axis and the lines $y = 0$ and $y = 3$.

271. The region is bounded by $y = e^x$, the y-axis and the line $y = e^2$. Find its exact area by integrating with respect to x.

272. Show that the area between $y = \ln x$, the x-axis and $x = e$ is 1 square unit, using the y-axis. (The rectangle $0 \le x \le e$, $0 \le y \le 1$ has area e.)

Tier 5 · rectangle minus the other side, and sign changes

273. Find the area bounded by $y = x^2$, $x = 2$ and the x-axis in two ways: by dx, and as a rectangle minus the area to the left found with dy.

274. The curve $y = x^3$ is drawn for $-2 \le x \le 2$. Find the total area between the curve and the y-axis from $y = -8$ to $y = 8$.

275. Find the area between the curve $y = x^2$ $(x \le 0)$, the y-axis and the lines $y = 1$ and $y = 9$.

Tier 6 · reflection of a^x and $\log_a x$

276. The curve $y = 2^x$ meets $y = 1$ at $x = 0$ and $y = 8$ at $x = 3$. Find the exact area between the curve, the y-axis and the lines $y = 1$ and $y = 8$.

277. Use the reflection in $y = x$ to show that the area between $y = \log_2 x$, the x-axis and $x = 8$ is $24 - \dfrac{7}{\ln 2}$.

278. Find the area between $y = \log_3 x$, the x-axis and $x = 9$ in exact form.

Conceptual

279. Explain why the limits are y-values when integrating $\displaystyle\int x\,dy$.

280. A student finds the area between $y = x^2$ $(x \ge 0)$, the y-axis and $y = 4$ as $\displaystyle\int_0^4 x^2 dx$. Explain the error and give the correct area.

281. Explain, using a sketch, why the area between $y = \ln x$ and the y-axis from $y = 0$ to $y = 2$ is $e^2 - 1$, while the area between $y = \ln x$ and the x-axis from $x = 1$ to $x = e^2$ is $e^2 + 1$.

282. Let R be the region bounded by $y = \sqrt{x}$, the y-axis and the line $y = 2$.

(a) Show that $x = y^2$. (1 mark)

(b) Find the area of R. (2 marks)

(c) Find the area between $y = \sqrt{x}$, the x-axis and $x = 4$, and check it with your answer to (b). (2 marks)

283. The area between $y = \sqrt{x}$, the y-axis and $y = k$ is 9 square units. Find $k > 0$.

Revision

284. Let $f(x) = x^2 - 2x - 3$.

(a) Find the x-intercepts. (1 mark)

(b) Find the area enclosed by the curve and the x-axis. (3 marks)

(c) Find the area enclosed by the curve, the y-axis and the line $y = 0$ for $x \geq 0$. (1 mark)

285. Solve $\log_2(x+1) + \log_2(x-1) = 3$.

(a) Show that $x^2 - 1 = 8$. (2 marks)

(b) Hence solve the equation. (2 marks)

286. Let $y = \dfrac{e^x}{x}$ for $x > 0$.

(a) Show that $\dfrac{dy}{dx} = \dfrac{e^x(x-1)}{x^2}$. (2 marks)

(b) Find the stationary point and its nature. (2 marks)

(c) Evaluate $\displaystyle\int_0^1 e^{2x}dx$. (1 mark)

12 The trapezoidal rule and rectangles

ANCHOR QUESTION

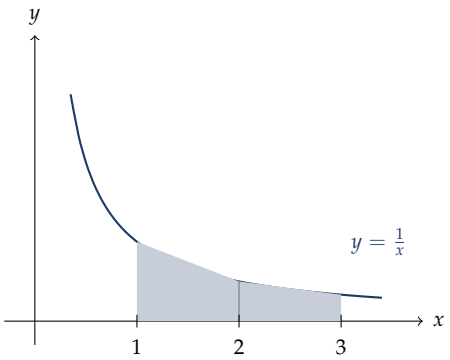
Use the trapezoidal rule with 4 strips to estimate $\displaystyle\int_0^2 \frac{1}{1+x^2}\,dx$.

SYLLABUS MAV-12-05

- Use a graphing application to compare different methods of approximating the area of the region between $y = f(x)$ and the x-axis, where $f(x) \geq 0$, by summing the areas of trapezia or rectangles each of width $\Delta x = \dfrac{b-a}{n}$, and observe the effect on the precision of the approximation as n increases
- Use the Trapezoidal rule to approximate integrals

NOTES

- One trapezium: $\displaystyle\int_a^b f(x)\,dx \approx \frac{b-a}{2}\Big[f(a)+f(b)\Big]$.
- n strips: $\displaystyle\int_a^b f(x)\,dx \approx \frac{h}{2}\Big[f(x_0)+2\big(f(x_1)+\cdots+f(x_{n-1})\big)+f(x_n)\Big]$, where $h = \dfrac{b-a}{n}$. This is on the reference sheet.
- List the x-values $x_k = a + kh$ first, then the function values, then use the rule. 'Two applications' means $n = 2$.
- Concave up ($f'' > 0$): the chords lie above the curve, so the estimate is too large. Concave down ($f'' < 0$): too small.
- Rectangles of width h: the left sum uses $f(x_0), \ldots, f(x_{n-1})$ and the right sum uses $f(x_1), \ldots, f(x_n)$. For an increasing function the left sum is an under-estimate and the right sum an over-estimate; for a decreasing function the two swap.
- Increasing n makes the estimate more precise.



WORKED EXAMPLE

Use the trapezoidal rule with 4 strips to estimate $\displaystyle\int_1^3 \ln x\,dx$, *to 3 decimal places.*

$h = \dfrac{3-1}{4} = 0.5$, so $x = 1,\ 1.5,\ 2,\ 2.5,\ 3$. list the x-values

$\ln x$: $0,\ 0.4055,\ 0.6931,\ 0.9163,\ 1.0986$

$\dfrac{0.5}{2}\Big[0 + 2(0.4055 + 0.6931 + 0.9163) + 1.0986\Big] \approx 1.282$

Tier 1 · h and the x-values

287. Find h and list the x-values.

(a) $[0,2]$, $n=4$

(b) $[1,3]$, $n=2$

(c) $[0,1]$, $n=4$

288. Find h and list the x-values.

(a) $[0,3]$, $n=3$

(b) $[2,6]$, $n=4$

(c) $[1,2]$, $n=5$

289. Why is $h=\dfrac{b-a}{n}$ and not $\dfrac{b-a}{n+1}$?

Tier 2 · one trapezium

290. Estimate with one trapezium.

(a) $\displaystyle\int_0^2 x^2\,dx$

(b) $\displaystyle\int_1^3 \frac{1}{x}\,dx$

291. Estimate with one trapezium.

(a) $\displaystyle\int_0^4 \sqrt{x}\,dx$

(b) $\displaystyle\int_0^2 2^x\,dx$

(c) $\displaystyle\int_1^2 x^3\,dx$

292. Show that one trapezium gives the exact value of $\displaystyle\int_1^4 (2x+1)\,dx$.

Tier 3 · tables of values

293. Use the trapezoidal rule on the table to estimate $\displaystyle\int_0^4 f(x)\,dx$.

x	0	1	2	3	4
$f(x)$	3	5	6	4	2

294. Use the trapezoidal rule on the table to estimate $\displaystyle\int_0^9 g(x)\,dx$.

x	0	3	6	9
$g(x)$	1	4	6	5

295. Use the trapezoidal rule with all six values to estimate $\int_0^5 g(t)\,dt$.

t	0	1	2	3	4	5
$g(t)$	60	55	65	68	70	67

Tier 4 · estimating from a function

296. Use two applications of the trapezoidal rule to estimate $\int_0^2 \sqrt{1+x^2}\,dx$ to 2 decimal places.

297. Use two applications of the trapezoidal rule to estimate $\int_0^1 e^x dx$ to 3 decimal places. Compare with the exact value.

298. Use two applications of the trapezoidal rule to estimate $\int_0^2 \left(\frac{1}{3}\right)^x dx$.

Tier 5 · over-estimate or under-estimate?

299. State whether the trapezoidal estimate is an over-estimate or an under-estimate. Use f''.

(a) $y = x^2$ on $[0,2]$

(b) $y = \sqrt{x}$ on $[0,4]$

(c) $y = e^x$ on $[0,1]$

300. Use two trapezia to estimate $\int_0^4 \sqrt{x}\,dx$, state whether it is too large or too small, and check with the exact value.

301. Show that $f(x) = e^{x^2}$ is concave up for all x, and state what this means for a trapezoidal estimate of $\int_0^1 e^{x^2} dx$.

Tier 6 · rectangles give bounds

302. For $f(x) = x^2$ on $[0,2]$ with 4 rectangles of equal width, find the left sum and the right sum, and show the exact area lies between them.

303. Use 4 rectangles of equal width to find a lower and an upper bound for the area under $y = \dfrac{1}{x}$ from $x = 1$ to $x = 3$.

304. For $y = e^x$ on $[0,1]$ with 4 rectangles, find a lower and an upper bound for the area, to 3 decimal places.

Conceptual

305. Show that the trapezoidal rule with 2 strips gives $\frac{h}{2}\left[f(x_0)+2f(x_1)+f(x_2)\right]$.

306. Explain what happens to the trapezoidal estimate of $\int_0^2 x^2 dx$ as n increases. Support with $n=1$ and $n=2$.

307. The shaded region is bounded by $y=\left(\frac{1}{3}\right)^x$, the coordinate axes and $x=2$.

(a) Use two applications of the trapezoidal rule to estimate its area. (2 marks)

(b) Show that the exact area is $\frac{8}{9\ln 3}$. (2 marks)

(c) Explain why the estimate in (a) is too large. (2 marks)

308. What is the trapezoidal estimate, using 2 strips, of $\int_0^4 f(x)\,dx$ if $f(0)=1$, $f(2)=5$ and $f(4)=3$?

A. 12 B. 14 C. 16 D. 28

Revision

309. Let $f(x)=3x^2-12$.

(a) Find the x-intercepts. (1 mark)

(b) Find the area between the curve and the x-axis from $x=-2$ to $x=2$. (2 marks)

(c) Find $\int_2^3 f(x)\,dx$. (2 marks)

310. Differentiate $\frac{x}{e^x}$ and hence evaluate $\int_0^1 \frac{x-1}{e^x}\,dx$.

(a) Differentiate xe^{-x}. (2 marks)

(b) Hence evaluate $\int_0^1 \frac{x-1}{e^x}\,dx$. (2 marks)

311. A sequence is defined by $T_n=5\times\left(\frac{1}{2}\right)^{n-1}$.

(a) Show that the sequence is geometric. (1 mark)

(b) Find the limiting sum. (2 marks)

(c) Find the least n for which $T_n<0.01$. (2 marks)

13 Integrals and inequalities

ANCHOR QUESTION

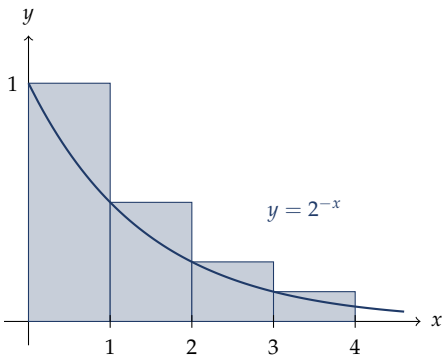
Rectangles of width 1 *lie above the curve* $y = 6^{-x}$ *at* $x = 0, 1, 2, \ldots$ *Show that the limiting sum of their areas is* $\dfrac{6}{5}$*, that* $\displaystyle\int_0^1 6^{-x}dx = \frac{5}{6\ln 6}$*, and hence that* $e^{25} < 6^{36}$.

SYLLABUS MAV-12-05

- Consider the definite integral as $\displaystyle\int_a^b f(x)\,dx = \lim_{\Delta x \to 0} \sum_a^b f(x)\,\Delta x$
- Apply $\displaystyle\int_a^b f(x)\,dx = F(b) - F(a)$, where $F(x)$ is a primitive of $f(x)$, to calculate definite integrals and solve related theoretical problems involving functions within the scope of the Mathematics Advanced course
- Use the Trapezoidal rule to approximate integrals

NOTES

- For a positive decreasing f: $f(1) + \cdots + f(n) < \displaystyle\int_0^n f(x)\,dx < f(0) + \cdots + f(n-1)$. Right-end rectangles sit under the curve; left-end rectangles stick out above it.
- Rectangles of width 1 under $y = b^{-x}$ $(b > 1)$ have areas $1, \frac{1}{b}, \frac{1}{b^2}, \ldots$ The limiting sum is $\dfrac{1}{1 - \frac{1}{b}} = \dfrac{b}{b-1}$.
- $\displaystyle\int_0^N b^{-x}dx = \left[-\frac{b^{-x}}{\ln b}\right]_0^N = \frac{1 - b^{-N}}{\ln b}$.
- Compare the integral with the sum to get an inequality involving $\ln b$. Multiply to clear fractions, then write $m \ln b = \ln b^m$ and use e^x is increasing: $\ln A > c$ means $A > e^c$.
- The same idea bounds $\ln n = \displaystyle\int_1^n \frac{1}{x}\,dx$ between sums of $\dfrac{1}{k}$.
- A concave-up curve is over-estimated by the trapezoidal rule, which also bounds integrals: for $f(x) = \dfrac{1}{x}$, $\ln n <$ the trapezoidal estimate.



WORKED EXAMPLE

Rectangles of width 1 *lie above the curve* $y = 2^{-x}$ *at* $x = 0, 1, 2, \ldots$ *Show that* $e^3 < 2^8$.

Areas $1, \frac{1}{2}, \frac{1}{4}, \ldots$: GP with $a = 1$, $r = \frac{1}{2}$, so the limiting sum is $\dfrac{1}{1 - \frac{1}{2}} = 2$.

$$\int_0^2 2^{-x}dx = \left[-\frac{2^{-x}}{\ln 2}\right]_0^2 = \frac{1}{\ln 2} - \frac{1}{4\ln 2} = \frac{3}{4\ln 2}$$

The integral is less than the sum of all the rectangles: $\dfrac{3}{4\ln 2} < 2$, so $3 < 8\ln 2 = \ln 2^8$ and $e^3 < 2^8$.

Tier 1 · limiting sums

312. Rectangles of width 1 and heights $y = b^{-x}$ at $x = 0, 1, 2, \ldots$ have areas forming a geometric series. Find the limiting sum when:

(a) $b = 2$

(b) $b = 4$

(c) $b = 10$

313. Show that the limiting sum for $y = b^{-x}$ is $\dfrac{b}{b-1}$ for $b > 1$.

314. Why must $b > 1$ for this series to have a limiting sum?

Tier 2 · the integral $\int_0^N b^{-x}dx$

315. Show that $\displaystyle\int_0^2 3^{-x}dx = \frac{8}{9\ln 3}$.

316. Show that $\displaystyle\int_0^3 2^{-x}dx = \frac{7}{8\ln 2}$ and $\displaystyle\int_0^1 5^{-x}dx = \frac{4}{5\ln 5}$.

317. Show that $\displaystyle\int_0^4 2^{-x}dx = \frac{15}{16\ln 2}$.

Tier 3 · from ln to powers

318. Rewrite as an inequality about powers of e.

(a) $\ln 2 > \dfrac{15}{32}$

(b) $\ln 3 > \dfrac{16}{27}$

(c) $\ln 5 < \dfrac{7}{4}$

319. Rewrite as an inequality about powers of e.

(a) $\ln 2 > \dfrac{3}{8}$

(b) $\ln 4 > \dfrac{7}{16}$

(c) $\ln 3 < \dfrac{4}{3}$

(d) $\ln 7 > \dfrac{5}{4}$

320. Check numerically that $2^8 > e^3$ and $3^{27} > e^{16}$.

Tier 4 · the full argument

321. Rectangles of width 1 lie above $y = 3^{-x}$ at $x = 0, 1, 2, \ldots$

(a) Show that the limiting sum of their areas is $\dfrac{3}{2}$. (1 mark)

(b) Show that $\displaystyle\int_0^1 3^{-x}dx = \frac{2}{3\ln 3}$. (2 marks)

(c) Hence show that $e^4 < 3^9$. (2 marks)

322. Rectangles of width 1 lie above $y = 4^{-x}$ at $x = 0, 1, 2, \ldots$ Show that $e^{189} < 4^{256}$.

323. Use $y = 10^{-x}$ and rectangles of width 1 to show that $e^{81} < 10^{100}$.

Tier 5 · bounding $\ln n$ with rectangles

324. Use rectangles of width 1 to show that $\dfrac{13}{12} < \ln 4 < \dfrac{11}{6}$.

325. Use rectangles of width 1 to show that $\dfrac{5}{6} < \ln 3 < \dfrac{3}{2}$.

326. Show that $\dfrac{77}{60} < \ln 5 < \dfrac{25}{12}$ and use the lower bound to show $5^{60} > e^{77}$.

Tier 6 · trapezia as an upper bound

327. Use the trapezoidal rule with one strip on $\displaystyle\int_1^3 \frac{1}{x}\,dx$ to show that $\ln 3 < \frac{4}{3}$, and explain why the estimate is an over-estimate.

328. Use one trapezium on $\displaystyle\int_1^2 \frac{1}{x}\,dx$ to show that $2^4 < e^3$.

329. Use two trapezia on $\displaystyle\int_1^3 \frac{1}{x}\,dx$ to show that $3^6 < e^7$.

Conceptual

330. Prove that for a positive decreasing function f, $\displaystyle\int_0^n f(x)\,dx < f(0)+f(1)+\cdots+f(n-1)$.

331. Explain why $\ln A > c$ implies $A > e^c$.

332. A student concludes $e^{20} < 2^{29}$ from $\displaystyle\int_0^4 2^{-x}dx < 2$. Explain what the inequality really proves.

333. Rectangles of width 1 lie above $y = 7^{-x}$ at $x = 0, 1, 2, \ldots$

(a) Show that the limiting sum of their areas is $\frac{7}{6}$. (1 mark)

(b) Show that $\displaystyle\int_0^1 7^{-x}dx = \frac{6}{7\ln 7}$. (2 marks)

(c) Hence show that $e^{36} < 7^{49}$. (2 marks)

334. Which is correct for $f(x) = \dfrac{1}{x}$ on $[1,4]$ with rectangles of width 1?

A. $\frac{1}{2}+\frac{1}{3}+\frac{1}{4} < \ln 4 < 1+\frac{1}{2}+\frac{1}{3}$ B. $1+\frac{1}{2}+\frac{1}{3} < \ln 4 < \frac{1}{2}+\frac{1}{3}+\frac{1}{4}$ C. $\ln 4 < \frac{1}{2}+\frac{1}{3}+\frac{1}{4}$ D. $\ln 4 > 1+\frac{1}{2}+\frac{1}{3}$

Revision

335. Consider $\displaystyle\int_1^e \frac{1}{x}\,dx$.

(a) Evaluate the integral. (1 mark)

(b) Use the trapezoidal rule with one strip to estimate it, to 3 decimal places. (2 marks)

(c) Explain why the estimate is greater than the exact value. (1 mark)

336. A curve has gradient $\dfrac{dy}{dx} = \dfrac{2x}{x^2+1} + \cos 2x$ and passes through $(0,3)$.

(a) Find the equation of the curve. (3 marks)

(b) Find the exact y-value when $x = \dfrac{\pi}{4}$. (1 mark)

337. A fair coin is tossed four times.

(a) Find the probability of exactly two heads. (2 marks)

(b) Find the probability of at least one head. (1 mark)

(c) Find the probability that the first and last tosses are both heads. (1 mark)

Solutions

1. (a) $2+3+4+5=14$ (b) $3+4+5+6=18$ (c) Trapezium: $\frac{1}{2}\times 4\times(2+6)=16$, between 14 and 18.

2. Left: $0+1+4=5$. Right: $1+4+9=14$.

3. (a) Left: $0.5(0+1+2+3)=3$. Right: $0.5(1+2+3+4)=5$. (b) $\frac{1}{2}\times 2\times 4=4$, which lies between 3 and 5.

4. Rectangle, width 4, height 3: 12.

5. Triangle base 4, height 2: 4. Trapezium height 6, parallel sides 1 and 7: $\frac{1}{2}\times 6\times 8=24$.

6. At $x=2$, $y=1$; at $x=5$, $y=7$. $\frac{1}{2}\times 3\times(1+7)=12$.

7. Rectangle below the axis, area 8, so -8.

8. Triangle below the axis, base 3, height 3: area 4.5, so -4.5.

9. Parallel sides 1 and 4, height 3: area 7.5, so -7.5.

10. Crosses at $x=2$. Below: triangle base 2, height 2: -2. Above: triangle base 4, height 4: 8. Total 6.

11. (a) Crosses at $x=2$. Below: base 3, height 6: -9. Above: base 3, height 6: 9. Integral $=0$. (b) $9+9=18$.

12. Crosses at $x=4$. Above: base 4, height 4: 8. Below: base 3, height 3: 4.5. Integral $=8-4.5=3.5$. Total area $=12.5$.

13. Upper semicircle, radius 2: $\frac{1}{2}\pi(2)^2=2\pi$.

14. Lower semicircle, radius 5: area $\frac{25\pi}{2}$, below the axis, so $-\frac{25\pi}{2}$.

15. Rectangle $6\times 2=12$ plus semicircle $\frac{1}{2}\pi(3)^2=\frac{9\pi}{2}$. Total $12+\frac{9\pi}{2}$.

16. In the sum $\sum f(x)\Delta x$, a height $f(x)$ below the axis is negative, so the integral subtracts that region. An area is a size and is always positive.

17. No. For $f(x)=x-2$, the region above the axis from 2 to 4 has the same area as the region below from 0 to 2, so the integral is 0.

18. $y=x$ is odd. The triangle from -3 to 0 is below the axis and the triangle from 0 to 3 is above, with equal areas 4.5, so they cancel.

19. The left sum increases and the right sum decreases, and both approach the definite integral.

20. (a) 4 (b) Above from 2 to 4: 2. Below from 4 to 6: -2. Integral $=0$. (c) $4+2+2=8$

21. (a) $x=-\dfrac{-8}{2\times 2}=2$ (b) $\Delta=64-8k=0$, so $k=8$. (c) $2x^2-8x+6=0$, so $x^2-4x+3=0$ and $x=1$ or $x=3$.

22. (a) $6\times\dfrac{\pi}{3}=2\pi$ cm (b) $\frac{1}{2}\times 36\times\dfrac{\pi}{3}=6\pi$ cm^2 (c) Triangle OAB: $\frac{1}{2}\times 6\times 6\times\sin\dfrac{\pi}{3}=9\sqrt{3}$. Segment $=6\pi-9\sqrt{3}$ cm^2.

23. (a) Product rule: $f'(x) = e^{2x} + x \times 2e^{2x} = e^{2x}(1+2x)$. (b) $1+2x=0$, so $x=-\frac{1}{2}$ and $y=-\frac{1}{2}e^{-1}$. Point $\left(-\frac{1}{2}, -\frac{1}{2e}\right)$. (c) $f''(x) = e^{2x}(4+4x)$. At $x=-\frac{1}{2}$, $f''=2e^{-1}>0$, so a minimum.

24. (a) $\dfrac{x^6}{6}+C$ (b) x^4+C (c) $x^{10}+C$ (d) $7x+C$

25. $\dfrac{d}{dx}\dfrac{x^6}{6} = \dfrac{6x^5}{6} = x^5$. General primitive: $\dfrac{x^6}{6}+C$.

26. $2x^3+C$, $\quad \dfrac{x^2}{2}+C$, $\quad x+C$

27. $3x^4+3x^2-7x+C$

28. $\dfrac{x^5}{5}+\dfrac{x^4}{2}-\dfrac{x^2}{2}+C$

29. $\dfrac{x^3}{9}-\dfrac{5x^2}{2}+2x+C$

30. $\dfrac{x^4}{4}+\dfrac{2}{x^2}+C$

31. $\frac{2}{3}x^{\frac{3}{2}}+2\sqrt{x}+C$

32. $-\dfrac{1}{x}+C$ and $\dfrac{9}{2}x^{\frac{2}{3}}+C$

33. $2x^2+x-15$, so $\dfrac{2x^3}{3}+\dfrac{x^2}{2}-15x+C$

34. $\displaystyle\int \left(x^2-3+2x^{-2}\right)dx = \frac{x^3}{3}-3x-\frac{2}{x}+C$

35. $x+4x^{\frac{1}{2}}+4$, so $\dfrac{x^2}{2}+\frac{8}{3}x^{\frac{3}{2}}+4x+C$

36. (a) $\dfrac{(2x+1)^4}{8}+C$ (b) $\dfrac{(3x-2)^6}{18}+C$ (c) $-\dfrac{1}{x+4}+C$

37. $\dfrac{(4x+1)^{\frac{3}{2}}}{4\times\frac{3}{2}}+C = \dfrac{(4x+1)^{\frac{3}{2}}}{6}+C$

38. $6\times\dfrac{(3x-1)^{-1}}{3\times(-1)}+C = -\dfrac{2}{3x-1}+C$

39. $f(x)=2x^3-2x+C$. $5=0+C$, so $f(x)=2x^3-2x+5$.

40. $f(x)=2x^{\frac{3}{2}}+C$. $10=16+C$, so $C=-6$ and $f(x)=2x^{\frac{3}{2}}-6$.

41. $y=(2x-1)^4+C$. $3=1+C$, so $y=(2x-1)^4+2$.

42. $\dfrac{d}{dx}\left(\dfrac{x^{n+1}}{n+1}\right) = \dfrac{(n+1)x^n}{n+1} = x^n$. The condition $n\neq -1$ keeps the denominator non-zero.

43. The denominator becomes $-1+1=0$, so the formula is undefined. A different primitive is needed (the natural logarithm, in a later episode).

44. $\frac{d}{dx}(x^2+1)^2 = 2(x^2+1)\times 2x = 4x^3+4x$ and $\frac{d}{dx}(x^4+2x^2) = 4x^3+4x$. They differ by 1, a constant, since $(x^2+1)^2 = x^4+2x^2+1$.

45. (a) $y = 2x^3-2x^2+C$; $3 = 0+C$, so $y = 2x^3-2x^2+3$. (b) $6x^2-4x = 2x(3x-2) = 0$. $y(0)=3$ and $y\left(\frac{2}{3}\right) = \frac{16}{27}-\frac{24}{27}+3 = \frac{73}{27}$.

46. B. Divide by $a(n+1) = 3\times 5 = 15$. cf. 2024 HSC Q5

47. (a) $5d = 15$, so $d=3$. (b) $a+2\times 3 = 11$, so $a=5$. (c) $S_{20} = \frac{20}{2}\big(2\times 5+19\times 3\big) = 10\times 67 = 670$.

48. (a) $2x-4=0$, so $x=2$. (b) Below: triangle base 2, height 4: -4. Above: triangle base 4, height 8: 16. Integral $=12$. (c) $4+16=20$.

49. (a) Let $u=2^x$: $u^2-5u+4=(u-1)(u-4)=0$. (b) $x=0$ or $x=2$. (c) $\log_3 27 = 3$.

50. $2x^3-x^2+C$. Then $\Big[2x^3-x^2\Big]_1^2 = (16-4)-(2-1) = 11$.

51. (a) $\left[\frac{x^3}{3}\right]_0^3 = 9$ (b) $\left[\frac{x^4}{4}\right]_1^2 = 4-\frac{1}{4} = \frac{15}{4}$ (c) $\Big[4x\Big]_0^5 = 20$

52. With $F(x)+C$, $\big(F(b)+C\big)-\big(F(a)+C\big) = F(b)-F(a)$: the C cancels.

53. $\Big[x^3+x\Big]_0^2 = 10$.

54. $\Big[x^4-x^2\Big]_1^3 = (81-9)-(1-1) = 72$.

55. $\left[\frac{x^3}{3}-\frac{x^2}{2}+3x\right]_{-2}^1 = \left(\frac{1}{3}-\frac{1}{2}+3\right)-\left(-\frac{8}{3}-2-6\right) = \frac{17}{6}+\frac{32}{3} = \frac{27}{2}$.

56. $\left[\frac{2}{3}x^{\frac{3}{2}}\right]_1^4 = \frac{2}{3}(8-1) = \frac{14}{3}$.

57. $\left[-\frac{1}{x}\right]_1^2 = -\frac{1}{2}+1 = \frac{1}{2}$.

58. $\left[2\sqrt{x}+\frac{x^2}{2}\right]_1^4 = (4+8)-\left(2+\frac{1}{2}\right) = \frac{19}{2}$.

59. $\left[\frac{(2x+1)^4}{8}\right]_0^1 = \frac{81-1}{8} = 10$.

60. $\left[-\frac{1}{x+1}\right]_1^3 = -\frac{1}{4}+\frac{1}{2} = \frac{1}{4}$.

61. $\left[\frac{(2x+1)^{\frac{3}{2}}}{3}\right]_0^2 = \frac{5^{\frac{3}{2}}-1}{3} = \frac{5\sqrt{5}-1}{3}$.

62. (a) $12-7=5$ (b) -12 (c) $2\times 7+3=17$

63. $F(2)-F(2)=0$. Also $\int_1^3 x^2dx = 9-\frac{1}{3}=\frac{26}{3}$ and $\int_3^1 x^2dx=\frac{1}{3}-9=-\frac{26}{3}$.

64. $\int_0^4 x^2dx=\dfrac{64}{3}$.

65. Both terms are odd, so the integral is 0.

66. $2\left[\dfrac{x^5}{5}\right]_0^2=\dfrac{64}{5}$.

67. x^5 is odd and integrates to 0. $3x^2$ is even: $2\int_0^1 3x^2dx=2$.

68. (a) $\left[t^3-2t\right]_1^x = x^3-2x-(1-2)=x^3-2x+1$ (b) $A'(x)=3x^2-2$ and $A(1)=0$.

69. $A(x)=2x^2-\dfrac{x^3}{3}$. $A'(x)=4x-x^2=x(4-x)=0$ at $x=0$ and $x=4$.

70. By the Fundamental Theorem, $\sqrt{x^2+1}$.

71. $A(x+h)-A(x)$ is the thin strip of area from x to $x+h$, about $f(x)\times h$. Dividing by h and letting $h\to 0$ gives $A'(x)=f(x)$, so A is a primitive of f.

72. $a^2+a=6$, so $(a+3)(a-2)=0$ and $a=2$ since $a>0$.

73. $\left[\dfrac{x^6}{6}\right]_{-a}^{a}=\dfrac{a^6}{6}-\dfrac{a^6}{6}=0$. x^5 is odd, so the signed areas on each side of the y-axis cancel.

74. (a) $A(x)=\dfrac{x^3}{3}-\dfrac{3x^2}{2}$. (b) $x^2\left(\dfrac{x}{3}-\dfrac{3}{2}\right)=0$, so $x=\dfrac{9}{2}$. (c) $A'(x)=x(x-3)$ changes from negative to positive at $x=3$.

75. A. The derivative of the accumulated area is the function itself. cf. 2021 HSC Q15

76. (a) $f'(x)=3x^2-6x=3x(x-2)=0$, so $x=0$ or $x=2$. (b) $f''(x)=6x-6$. $f''(0)=-6<0$, a maximum. $f''(2)=6>0$, a minimum. (c) $f(2)=8-12+k=-1$, so $k=3$.

77. (a) $\dfrac{4}{7}\times\dfrac{3}{6}=\dfrac{2}{7}$ (b) $\dfrac{4}{7}\times\dfrac{3}{6}+\dfrac{3}{7}\times\dfrac{4}{6}=\dfrac{24}{42}=\dfrac{4}{7}$ (c) $\dfrac{\frac{2}{7}}{\frac{4}{7}}=\dfrac{1}{2}$

78. (a) $x=\dfrac{\pi}{3}$ or $\dfrac{2\pi}{3}$ (b) $\cos x(2\cos x-1)=0$: $x=\dfrac{\pi}{2},\ \dfrac{3\pi}{2},\ \dfrac{\pi}{3},\ \dfrac{5\pi}{3}$

79. (a) $f=x^2+4$, $f'=2x$, $k=\frac{1}{2}$ (b) $f=x^3-1$, $f'=3x^2$, $k=2$ (c) $f=x^2+3x$, $f'=2x+3$, $k=1$

80. (a) Yes: $f=x^3+5$, $f'=3x^2$. (b) No: x is not a multiple of $3x^2$. (c) No: there is no factor $2x$ outside.

81. $f'(x)=15x^2-2$, so $15x^2-2=f'(x)$ exactly.

82. $\dfrac{\left(x^2+1\right)^5}{5}+C$

83. $\dfrac{\left(x^3-2\right)^4}{4}+C$

84. $\dfrac{\left(x^2+3x\right)^3}{3}+C$

85. $f'=2x$, so $\frac{1}{2}\times\dfrac{\left(x^2+4\right)^4}{4}+C=\dfrac{\left(x^2+4\right)^4}{8}+C$

86. $\dfrac{\left(x^3+1\right)^3}{9}+C$

87. $f'=-2x$, so $-\frac{1}{2}\times\dfrac{\left(1-x^2\right)^6}{6}+C=-\dfrac{\left(1-x^2\right)^6}{12}+C$

88. $\frac{1}{2}\int 2x\left(x^2+9\right)^{-\frac{1}{2}}dx=\frac{1}{2}\times 2\left(x^2+9\right)^{\frac{1}{2}}+C=\sqrt{x^2+9}+C$

89. $\frac{1}{2}\int 2x\left(x^2+1\right)^{-2}dx=-\dfrac{1}{2\left(x^2+1\right)}+C$

90. $\frac{1}{3}\int 3x^2\left(x^3+1\right)^{\frac{1}{2}}dx=\dfrac{2}{9}\left(x^3+1\right)^{\frac{3}{2}}+C$

91. $\left[\dfrac{\left(x^2+1\right)^4}{8}\right]_0^1=\dfrac{16-1}{8}=\dfrac{15}{8}$

92. $\left[\sqrt{x^2+5}\right]_0^2=3-\sqrt{5}$

93. $\left[-\dfrac{1}{x^3+2}\right]_1^2=-\frac{1}{10}+\frac{1}{3}=\frac{7}{30}$

94. $f'=2x+2=2(x+1)$, so $\dfrac{\left(x^2+2x\right)^4}{8}+C$

95. $\int(2x+1)\left(x^2+x\right)^{-3}dx=-\dfrac{1}{2\left(x^2+x\right)^2}+C$

96. $f'=4x^3-4x=4\left(x^3-x\right)$, so $\dfrac{\left(x^4-2x^2\right)^6}{24}+C$

97. $\dfrac{d}{dx}\left(\dfrac{[f(x)]^{n+1}}{n+1}\right)=\dfrac{(n+1)[f(x)]^n f'(x)}{n+1}=f'(x)[f(x)]^n.$

98. There is no factor $2x$ (the derivative of the bracket) outside it. The method needs $f'(x)$ as a factor. Expanding the bracket would be the way to proceed.

99. A variable factor cannot be moved outside the integral. Correct: $\frac{1}{2}\times\dfrac{\left(x^2+1\right)^4}{4}+C=\dfrac{\left(x^2+1\right)^4}{8}+C$.

100. (a) $f=x^2-1$, $f'=2x$, so $6xf^2=3f'f^2$ and $y=\left(x^2-1\right)^3+C$. $2=0+C$, so $y=\left(x^2-1\right)^3+2$. (b) $6x(x^2-1)^2=0$ at $x=0,\pm1$. The gradient changes from negative to positive at $x=0$: minimum $(0,1)$. At $x=\pm1$ the gradient does not change sign: horizontal points of inflection $(\pm1,2)$.

101. A. $f' = 6x$, so $\frac{1}{6} \times \frac{f^5}{5}$. cf. 2023 HSC Q17

102. (a) Chain rule: $\frac{f'(x)}{f(x)}$ form gives $\frac{2x}{x^2+1}$. (b) $f(1) = \ln 2$, $f'(1) = 1$. $y - \ln 2 = x - 1$, so $y = x - 1 + \ln 2$. (c) $f(-x) = \ln\left((-x)^2 + 1\right) = f(x)$.

103. (a) x^3 is odd. $2x^2 + 1$ is even. (b) $2\int_0^1 \left(2x^2+1\right) dx = 2\left(\frac{2}{3}+1\right) = \frac{10}{3}$.

104. (a) $3 \times 2^5 = 96$ (b) $\frac{3\left(2^8-1\right)}{2-1} = 765$ (c) $\frac{8}{1-\frac{1}{2}} = 16$

105. (a) $\frac{e^{4x}}{4} + C$ (b) $-e^{-x} + C$ (c) $2e^{\frac{x}{2}} + C$

106. (a) $-4e^{-2x} + C$ (b) $\frac{e^{3x}}{6} + C$

107. $\frac{d}{dx}\left(\frac{1}{5}e^{5x}\right) = \frac{1}{5} \times 5e^{5x} = e^{5x}$.

108. $\frac{3}{2}e^{2x+5} + C$

109. $\frac{2}{3}e^{3x+1} + e^{-x} + C$

110. $-\frac{5}{2}e^{1-2x} + C$

111. (a) $\frac{2^x}{\ln 2} + C$ (b) $\frac{5^x}{\ln 5} + C$ (c) $\frac{3 \times 4^x}{\ln 4} + C$

112. $\frac{3^x}{\ln 3} + \frac{x^2}{2} + C$ and $\frac{2^x}{\ln 2} - 2x + C$

113. $\frac{5^x}{\ln 5} - \frac{x^6}{6} + C$

114. $\int e^{-2x} dx = -\frac{1}{2}e^{-2x} + C$

115. Expand: $e^{2x} + 2 + e^{-2x}$. Integral $= \frac{1}{2}e^{2x} + 2x - \frac{1}{2}e^{-2x} + C$.

116. $e^x + e^{-x}$, so $e^x - e^{-x} + C$.

117. $\left[\frac{e^{2x}}{2} + x\right]_0^1 = \frac{e^2}{2} + 1 - \frac{1}{2} = \frac{e^2+1}{2}$

118. $\left[\frac{e^{2x}}{2}\right]_0^{\ln 2} = \frac{4}{2} - \frac{1}{2} = \frac{3}{2}$ (using $e^{2\ln 2} = 4$).

119. $\left[\frac{e^{2x+1}}{2}\right]_{-1}^1 = \frac{e^3 - e^{-1}}{2}$

120. (a) $e^{x^3}+C$ (b) $\frac{1}{2}e^{x^2}+C$ (c) $e^{x^2-x}+C$

121. $f'=3x^2$, so $\frac{1}{3}e^{x^3+1}+C$.

122. $\left[e^{x^2+x}\right]_0^1 = e^2-1$

123. $f=-x^2$, $f'=-2x$, so $\left[-\frac{1}{2}e^{-x^2}\right]_0^1 = \frac{1}{2}\left(1-e^{-1}\right)$

124. $\dfrac{d}{dx}\left(\dfrac{1}{a}e^{ax+b}\right) = \dfrac{1}{a}\times ae^{ax+b} = e^{ax+b}$. The condition $a\neq 0$ keeps $\dfrac{1}{a}$ defined.

125. e is the constant $2.718\ldots$, so $\int e\,dx$ is the integral of a constant, giving ex. The e^x rule needs x in the power.

126. B. cf. 2020 HSC Q4

127. $f(x)=2e^{3x}+C$. $5=2+C$, so $f(x)=2e^{3x}+3$.

128. (a) $\left[e^x-e^{-x}\right]_0^{\ln 2} = \left(2-\frac{1}{2}\right)-(1-1) = \frac{3}{2}$ (b) $f(-x)=e^{-x}+e^x=f(x)$. So the integral is $2\times\frac{3}{2}=3$.

129. (a) $y=x^3-3x^2+C$; $2=1-3+C$, so $C=4$: $y=x^3-3x^2+4$. (b) $3x^2-6x=0$ at $x=2$, and $y(2)=8-12+4=0$. (c) $\left[\dfrac{x^4}{4}-x^3+4x\right]_0^2 = 4-8+8=4$.

130. (a) $f'(x)=2xe^x+x^2e^x=xe^x(2+x)$. (b) $x=0$ gives $(0,0)$. $x=-2$ gives $\left(-2,\,4e^{-2}\right)$. (c) $f'(x)$ changes from negative (for $-2<x<0$) to positive: minimum.

131. (a) $x+3=16$, so $x=13$. (b) $2x=4$, so $x=2$. (c) $x(x-4)=5$, so $x^2-4x-5=0$ and $x=5$ (reject $x=-1$).

132. (a) $7\ln|x|+C$ (b) $\frac{1}{2}\ln|x|+C$ (c) $\ln|x|+\dfrac{x^2}{2}+C$

133. $\dfrac{d}{dx}\ln x=\dfrac{1}{x}$ for $x>0$.

134. $\left[3\ln x\right]_1^4 = 3\ln 4-3\ln 1=3\ln 4$

135. (a) $\frac{1}{3}\ln|3x+1|+C$ (b) $2\ln|3x-1|+C$ (c) $-\ln|1-x|+C$

136. $\frac{1}{2}\ln|4x-3|+\frac{1}{2}e^{2x}+C$

137. $\left[\frac{1}{2}\ln(2x+1)\right]_0^2 = \frac{1}{2}\ln 5$

138. (a) $f=x^2+4$, $f'=2x$, $k=\frac{1}{2}$ (b) $f=x^3+1$, $f'=3x^2$, $k=\frac{1}{3}$ (c) $f=x^2+3x$, $f'=2x+3$, $k=1$

139. $\ln\left(x^2+1\right)+C$ and $\frac{1}{2}\ln\left|x^2-9\right|+C$

140. $\frac{1}{6}\ln\left|2x^3+5\right|+C$

141. $\frac{1}{2}\ln\left(4+x^2\right)+C$ cf. 2020 HSC Q17

142. $\Big[2\ln\left(x^3+1\right)\Big]_0^1 = 2\ln 2$

143. $\Big[\frac{1}{2}\ln\left(x^2-1\right)\Big]_2^5 = \frac{1}{2}\ln 24 - \frac{1}{2}\ln 3 = \frac{1}{2}\ln 8$

144. $\int\left(x+3+\frac{1}{x}\right)dx = \frac{x^2}{2}+3x+\ln|x|+C$

145. $\left[\ln x - \frac{2}{x}\right]_1^e = \left(1-\frac{2}{e}\right)-(0-2) = 3-\frac{2}{e}$

146. $3\ln|2x+1|-3\ln|x|+2x^2+C$

147. $\frac{d}{dx}\ln|f(x)| = \frac{f'(x)}{f(x)}$ by the chain rule.

148. For $n=-1$ the formula divides by 0. The derivative of $\ln x$ is $\frac{1}{x}$, so $\ln|x|$ is the primitive.

149. $\frac{d}{dx}\ln\left(x^2+1\right) = \frac{2x}{x^2+1}$, not $\frac{1}{x^2+1}$. The top must be the derivative of the bottom.

150. (a) Integrate both sides: $\ln f(x) = x^2+C$ since $f(x)>0$. (b) $f(0)=1$ gives $C=0$, so $f(x)=e^{x^2}$.
cf. 2026 HSC Q31(b)

151. $f=1+\sin x$, $f'=\cos x$: $\Big[\ln(1+\sin x)\Big]_0^{\pi/2} = \ln 2-\ln 1 = \ln 2$.

152. (a) $f' = 2xe^{-x}-x^2e^{-x} = xe^{-x}(2-x)$. (b) $x=0$: $(0,0)$. $x=2$: $\left(2,\,4e^{-2}\right)$. (c) $\Big[e^x-e^{-x}\Big]_0^1 = e-e^{-1}$.

153. (a) $y = 2\ln(2x+1)+C$. $3=0+C$, so $y=2\ln(2x+1)+3$. (b) $y=2\ln 5+3$. (c) $\frac{dy}{dx} = \frac{4}{2x+1} > 0$ for $x>-\frac{1}{2}$.

154. (a) $5r^3=-40$, so $r^3=-8$ and $r=-2$. (b) $S_{10} = \frac{5\left(1-(-2)^{10}\right)}{1-(-2)} = \frac{5(1-1024)}{3} = -1705$. (c) $|r|=2>1$.

155. (a) $\sin x+C$ (b) $-\cos x+C$ (c) $\tan x+C$

156. $3\sin x+2\cos x+C$

157. $\frac{d}{dx}(-\cos x) = -(-\sin x) = \sin x$.

158. (a) $\frac{1}{4}\sin 4x+C$ (b) $-\frac{1}{2}\cos 2x+C$ (c) $\frac{1}{3}\tan 3x+C$

159. (a) $-2\cos(3x+1)+C$ (b) $-2\cos\frac{x}{2}+C$ (c) $\frac{1}{2}\sin\left(2x-\frac{\pi}{3}\right)+C$

160. $4\sin 2x + \frac{3}{4}\cos 4x + \tan x + C$

161. (a) 1 (b) $\Big[-\cos x\Big]_0^{\pi} = 1 + 1 = 2$ (c) $\tan\frac{\pi}{4} - \tan 0 = 1$

162. $\Big[\frac{1}{2}\sin 2x\Big]_0^{\pi/4} = \frac{1}{2}$ and $\Big[-\frac{1}{2}\cos 2x\Big]_0^{\pi/2} = 1$

163. $\tan\frac{\pi}{4} - \tan 0 = 1$ cf. 2020 HSC Q13

164. $\Big[\frac{1}{2}\sin 2x + \tan x\Big]_0^{\pi/4} = \frac{1}{2} + 1 = \frac{3}{2}$

165. $\Big[\sin 3x + \frac{1}{2}\tan 2x\Big]_0^{\pi/6} = 1 + \frac{\sqrt{3}}{2}$

166. $\Big[-\cos\left(x - \frac{\pi}{6}\right)\Big]_{\pi/6}^{\pi/2} = -\frac{1}{2} + 1 = \frac{1}{2}$

167. (a) $\frac{1}{2}\sin\left(x^2\right) + C$ (b) $-\cos\left(x^3\right) + C$ (c) $e^{\tan x} + C$

168. (a) $\dfrac{\sin^5 x}{5} + C$ (b) $\dfrac{\tan^4 x}{4} + C$ (c) $-\dfrac{\cos^3 x}{3} + C$

169. $\left[-\dfrac{\cos^3 x}{3}\right]_0^{\pi/2} = 0 + \dfrac{1}{3} = \dfrac{1}{3}$

170. $\ln|\sin x| + C$. Since $\tan x = \dfrac{\sin x}{\cos x} = -\dfrac{-\sin x}{\cos x}$, $\displaystyle\int \tan x\,dx = -\ln|\cos x| + C$.

171. $\Big[\ln\sin x\Big]_{\pi/6}^{\pi/2} = \ln 1 - \ln\frac{1}{2} = \ln 2$

172. $\Big[-\ln\cos x\Big]_0^{\pi/3} = -\ln\frac{1}{2} + \ln 1 = \ln 2$

173. $\dfrac{d}{dx}\left(-\dfrac{1}{a}\cos(ax+b)\right) = -\dfrac{1}{a} \times (-a\sin(ax+b)) = \sin(ax+b)$.

174. The region above the axis from 0 to π has area 2. The region below from π to 2π contributes -2. They cancel.

175. The integral divides by the coefficient 3: $\dfrac{1}{3}\sin 3x + C$. Check: $\dfrac{d}{dx}(3\sin 3x) = 9\cos 3x \neq \cos 3x$.

176. (a) $y = 3\cos 2x + C$. $5 = 3 + C$, so $y = 3\cos 2x + 2$. (b) $\cos 2x = 1$ gives $y = 5$ at $x = 0$; the next maximum is at $x = \pi$. The smallest positive x is π. cf. 2023 HSC Q26(a)

177. B.

178. (a) $\sin x + x\cos x$ (b) $y = \frac{\pi}{2}$ and gradient $= 1$. $y - \frac{\pi}{2} = x - \frac{\pi}{2}$, so $y = x$. (c) $\Big[\sin x\Big]_0^{\pi} = 0$.

179. (a) LHS $= \dfrac{1 - \cos^2\theta}{\cos\theta} = \dfrac{\sin^2\theta}{\cos\theta} = \sin\theta\tan\theta$. (b) $\sin\theta\,(\tan\theta - 1) = 0$: $\theta = 0,\ \pi,\ 2\pi$ or $\tan\theta = 1$: $\theta = \dfrac{\pi}{4},\ \dfrac{5\pi}{4}$.

180. (a) $\dfrac{\left(x^2+1\right)^4}{8}+C$ (b) $\ln\left(x^2+1\right)+C$ (c) $y=\ln\left(x^2+1\right)+2$, so $y=\ln 2+2$.

181. $xe^{3x}+C$

182. $x^2\sin x+C$

183. $\displaystyle\int 6x\left(x^2+1\right)^2dx=\left(x^2+1\right)^3$, so $\displaystyle\int x\left(x^2+1\right)^2dx=\frac{\left(x^2+1\right)^3}{6}+C$.

184. $\displaystyle\int 3xe^{3x}dx=xe^{3x}-\tfrac{1}{3}e^{3x}$, so $\displaystyle\int xe^{3x}dx=\frac{(3x-1)e^{3x}}{9}+C$.

185. $\displaystyle\int x\cos x\,dx=x\sin x-\int\sin x\,dx=x\sin x+\cos x+C$.

186. $x\ln x-x+C$

187. $\dfrac{d}{dx}(x\cos x)=\cos x-x\sin x$. So $x\cos x=\sin x-\displaystyle\int x\sin x\,dx$ and $\displaystyle\int x\sin x\,dx=\sin x-x\cos x+C$.

188. $\dfrac{d}{dx}(x\sin 2x)=\sin 2x+2x\cos 2x$. So $x\sin 2x=-\frac{1}{2}\cos 2x+2\displaystyle\int x\cos 2x\,dx$ and $\displaystyle\int x\cos 2x\,dx=\frac{1}{2}x\sin 2x+\frac{1}{4}\cos 2x+C$.

189. $\dfrac{d}{dx}\left(xe^{-2x}\right)=e^{-2x}-2xe^{-2x}$. So $xe^{-2x}=-\frac{1}{2}e^{-2x}-2\displaystyle\int xe^{-2x}dx$ and $\displaystyle\int xe^{-2x}dx=-\frac{1}{2}xe^{-2x}-\frac{1}{4}e^{-2x}+C$.

190. $\dfrac{d}{dx}(x\ln x)=\ln x+1$. So $x\ln x=\displaystyle\int\ln x\,dx+x$ and $\displaystyle\int\ln x\,dx=x\ln x-x+C$.

191. $\dfrac{d}{dx}\left(x^2\ln x\right)=2x\ln x+x^2\times\dfrac{1}{x}$. So $x^2\ln x=2\displaystyle\int x\ln x\,dx+\frac{x^2}{2}$ and $\displaystyle\int x\ln x\,dx=\frac{1}{2}x^2\ln x-\frac{1}{4}x^2+C$.

192. $\dfrac{d}{dx}\left(x^3\ln x\right)=3x^2\ln x+x^2$. So $\displaystyle\int x^2\ln x\,dx=\frac{x^3\ln x}{3}-\frac{x^3}{9}+C$.

193. $\dfrac{d}{dx}\left(x^2+3\right)^5=10x\left(x^2+3\right)^4$, so $\displaystyle\int x\left(x^2+3\right)^4dx=\frac{\left(x^2+3\right)^5}{10}+C$. cf. 2022 HSC Q18

194. $\dfrac{d}{dx}\left(e^{3x}(3x-1)\right)=3e^{3x}(3x-1)+3e^{3x}=9xe^{3x}$, so $\displaystyle\int xe^{3x}dx=\frac{e^{3x}(3x-1)}{9}+C$. cf. 2020 HSC Q18

195. $\dfrac{d}{dx}\dfrac{x}{x^2+1}=\dfrac{\left(x^2+1\right)-2x^2}{\left(x^2+1\right)^2}=\dfrac{1-x^2}{\left(x^2+1\right)^2}$, so the integral is $\dfrac{x}{x^2+1}+C$.

196. $\cos x-(\cos x-x\sin x)=x\sin x$. $\Big[\sin x-x\cos x\Big]_0^{\pi}=(0+\pi)-0=\pi$. cf. 2025 HSC Q25

197. $\sec^2 x-1=\tan^2 x$. $\Big[\tan x-x\Big]_0^{\pi/4}=1-\dfrac{\pi}{4}$.

198. $\dfrac{d}{dx}\left(e^{-x}(x+1)\right)=-e^{-x}(x+1)+e^{-x}=-xe^{-x}$. So $\displaystyle\int_0^1 xe^{-x}dx=\Big[-e^{-x}(x+1)\Big]_0^1=1-\frac{2}{e}$.

199. xe^{2x} is the primitive of the whole derivative $e^{2x} + 2xe^{2x}$, not of xe^{2x} alone. The extra e^{2x} term must be integrated and moved across: $\int xe^{2x}dx = \frac{1}{2}xe^{2x} - \frac{1}{4}e^{2x} + C$.

200. Use the result you have just found (here the derivative) to answer the next part, rather than starting a new method.

201. $\ln x + 1 - 1 = \ln x$. So $\displaystyle\int_1^e \ln x\,dx = \Big[x\ln x - x\Big]_1^e = (e-e) - (0-1) = 1$.

202. (a) Product rule: $2x\ln x + x^2 \times \dfrac{1}{x} = 2x\ln x + x$. (b) $\displaystyle\int_1^e (2x\ln x + x)dx = \Big[x^2\ln x\Big]_1^e = e^2$ and $\displaystyle\int_1^e x\,dx = \frac{e^2-1}{2}$. So $2\displaystyle\int_1^e x\ln x\,dx = e^2 - \frac{e^2-1}{2} = \frac{e^2+1}{2}$ and the integral is $\dfrac{e^2+1}{4}$. cf. 2024 HSC Q27(b)

203. $\sqrt{x+1} + \dfrac{x}{2\sqrt{x+1}} = \dfrac{2(x+1)+x}{2\sqrt{x+1}} = \dfrac{3x+2}{2\sqrt{x+1}}$.

204. (a) $f'(x) = 3x^2 - 12x + 9 = 3(x-1)(x-3)$, so $x = 1$ or $x = 3$. (b) $f(x) = x(x-3)^2 \geq 0$ for $x \geq 0$. (c) $\left[\dfrac{x^4}{4} - 2x^3 + \dfrac{9x^2}{2}\right]_0^3 = \dfrac{81}{4} - 54 + \dfrac{81}{2} = \dfrac{27}{4}$.

205. (a) $0.4 \times 0.5 = 0.2$ (b) $0.4 + 0.5 - 0.2 = 0.7$ (c) $P(A \mid B) = \dfrac{0.2}{0.5} = 0.4 = P(A)$. For independent events the occurrence of B does not change the probability of A.

206. (a) $\frac{3}{2}\ln(2x-1) - e^{-x} + C$ (b) $\left[\frac{3}{2}\ln(2x-1) - e^{-x}\right]_1^2 = \frac{3}{2}\ln 3 - e^{-2} + e^{-1}$.

207. (a) $\dfrac{1}{ax+b}$ (b) e^{ax+b} (c) $\dfrac{f'}{f}$ (d) $f'e^f$ (e) $f'f^n$

208. (a) $\sin(ax+b)$ (b) $\dfrac{f'}{f}$ (c) power, $n = -3$ (d) $(ax+b)^n$ (e) $\sec^2(ax+b)$

209. e^{3x}. The others need splitting, a negative power or expanding first.

210. $\frac{2}{3}\ln|3x+1| + \frac{1}{2}\cos 2x + C$

211. $\frac{1}{3}\tan 3x + 2\sqrt{x} + C$

212. $2\sin\dfrac{x}{2} + 2e^{\frac{x}{2}} - x^3 + C$

213. $\frac{x^2}{2} + 3x + \ln|x| + C$

214. $e^x + e^{-x}$ integrates to $e^x - e^{-x} + C$.

215. $x^2 - 2 + x^{-2}$, so $\dfrac{x^3}{3} - 2x - \dfrac{1}{x} + C$.

216. $e^{2x} + 2e^x + 1$, so $\dfrac{e^{2x}}{2} + 2e^x + x + C$.

217. $\frac{1}{3}e^{x^3} + \frac{1}{3}\ln\left|x^3+1\right| + C$

218. $\dfrac{(x^2-4)^4}{8} + C$ and $\dfrac{(\ln x)^3}{3} + C$

219. $\ln(e^x+1)+C$ and $-\ln(2+\cos x)+C$

220. $\frac{3}{2}$

221. $\left[\ln(x^4+1)-e^{2-x}\right]_1^2 = \ln\frac{17}{2}+e-1$

222. $1+\dfrac{e^2-1}{2}=\dfrac{e^2+1}{2}$ and $\dfrac{e^3+e^{-3}-2}{3}$

223. $y=3\ln(x^2+1)-\frac{1}{2}e^{-2x}+C$. $2=-\frac{1}{2}+C$, so $C=\frac{5}{2}$.

224. $f(x)=\frac{1}{2}\sin 2x-\dfrac{1}{x+1}+C$. $1=0-1+C$, so $C=2$.

225. $f(x)=\dfrac{2^x}{\ln 2}+C$. $3=\dfrac{1}{\ln 2}+C$, so $f(x)=\dfrac{2^x-1}{\ln 2}+3$.

226. The factor needed for $f'e^f$ is $2x$ and it is missing. No other form fits either.

227. The top must be the derivative of the bottom: $2x$. Here it is 1, so the ln rule does not apply.

228. A quotient cannot be integrated as numerator over denominator. Split first: $1+\dfrac{1}{x}$, so $x+\ln|x|+C$.

229. $\left[\ln(x^2+1)+\frac{1}{2}\ln(2x+1)\right]_0^1 = \ln 2+\frac{1}{2}\ln 3$.

230. A.

231. (a) Quotient rule: $\dfrac{\frac{1}{x}\times x-\ln x\times 1}{x^2}=\dfrac{1-\ln x}{x^2}$. (b) $\ln x=1$, so $x=e$ and $y=\dfrac{1}{e}$: $\left(e,\dfrac{1}{e}\right)$. (c) $\left[f(x)\right]_1^e=\dfrac{1}{e}-0=\dfrac{1}{e}$.

232. (a) $\sin x$ is odd. $\cos x$ is even. (b) $0+2\displaystyle\int_0^{\pi/3}\cos x\,dx=2\sin\frac{\pi}{3}=\sqrt{3}$.

233. (a) $(x-1)(x-3)=0$, so $x=1$ or 3. (b) $x=2$, $y=4-8+3=-1$: $(2,-1)$. (c) $x^2-4x=0$, so $x=0$ or 4. (d) $f(x)\geq -1$.

234. (a) Positive. Area 2. (b) Negative. Area 2. (c) Zero. Area 4.

235. (a) Positive. Area 4. (b) Negative. Area 3. (c) Zero. Area is not 0 (equal parts cancel).

236. False. It is true only when the curve stays on one side of the axis. If it crosses, positive and negative parts cancel inside the integral.

237. (a) $\left[x^3\right]_1^2=7$ (b) $3-1=2$ (c) $\ln 4$

238. (a) 1 (b) $\dfrac{e^2-1}{2}$ (c) $\left[2\ln(2x+1)\right]_0^2=2\ln 5$

239. $\displaystyle\int_{-3}^{3}\left(9-x^2\right)dx=36$

240. $\int_0^3 (-x^2)dx = -9$, so the area is 9.

241. Intercepts $x = 0, 6$. $\int_0^6 \left(x^2 - 6x\right) dx = 72 - 108 = -36$. Area $= 36$.

242. $\Big[-3\ln x\Big]_1^e = -3$, so the area is 3.

243. (a) $x = 3$ (and $x = 0$ at the end) (b) $x = 1$ (c) $x = \pi$ (plus the ends)

244. (a) $x = 2$ (b) $x = 3$ (c) $x = \dfrac{\pi}{2}$ (d) $x = \ln 2$

245. No. The curve crosses the axis at $x = 0$, so the integral is 0 (an odd function) but the area is positive.

246. $F(x) = \dfrac{x^4}{4} - 2x^3 + 4x^2$. $\int_0^2 = 4$ and $\int_2^4 = -4$. Area $= 8$.

247. $\int_0^3 = -\frac{9}{2}$ and $\int_3^5 = \frac{26}{3}$. Area $= \frac{9}{2} + \frac{26}{3} = \frac{79}{6}$.

248. Crosses at $x = \ln 2$. $\int_0^{\ln 2} = 1 - 2\ln 2 < 0$ and $\int_{\ln 2}^{\ln 4} = 2 - 2\ln 2$. Area $= (2\ln 2 - 1) + (2 - 2\ln 2) = 1$.

249. $\int_{-1}^0 = \frac{1}{4}$ and $\int_0^1 = -\frac{1}{4}$. Area $= \frac{1}{2}$.

250. $\int_0^1 2x\,dx = 1$ and $\int_1^3 \dfrac{2}{x}\,dx = 2\ln 3$. Area $= 1 + 2\ln 3$. cf. 2021 HSC Q24

251. $2\ln(2k+1) = 2\ln 5$, so $2k + 1 = 5$ and $k = 2$.

252. $9 - 3^x = 0$ at $x = 2$. $\int_0^2 (9 - 3^x)\,dx = \left[9x - \dfrac{3^x}{\ln 3}\right]_0^2 = \left(18 - \dfrac{9}{\ln 3}\right) + \dfrac{1}{\ln 3} = 18 - \dfrac{8}{\ln 3}$. cf. 2021 HSC Q28(a)

253. $\int_0^\pi \sin x\,dx = 2$ (above) and $\int_\pi^{2\pi} \sin x\,dx = -2$ (below). The integral adds $2 + (-2) = 0$. The area adds $2 + 2 = 4$.

254. The graph is odd, so the two areas are equal. Total $= 2\int_0^2 x^3 dx = 8$.

255. The curve crosses at $x = 2$. $\int_0^2 = -\frac{16}{3}$ and $\int_2^3 = \frac{7}{3}$. Area $= \frac{16}{3} + \frac{7}{3} = \frac{23}{3}$.

256. (a) $x = 0$ and $x = 4$. (b) -9 and $\frac{7}{3}$. (c) $9 + \frac{7}{3} = \frac{34}{3}$.

257. B. Subtract the (negative) second integral to make its area positive.

258. (a) $e^{2x} = 4$, so $2x = \ln 4$ and $x = \ln 2$. (b) $\left[\dfrac{e^{2x}}{2} - 4x\right]_0^{\ln 2} = (2 - 4\ln 2) - \dfrac{1}{2} = \dfrac{3}{2} - 4\ln 2$. (c) $f < 0$ there, so area $= 4\ln 2 - \dfrac{3}{2}$.

259. (a) $c^2 = 49 + 25 - 2 \times 7 \times 5 \times \frac{1}{2} = 39$, so $c = \sqrt{39}$. (b) $\frac{1}{2} \times 7 \times 5 \times \sin 60^\circ = \dfrac{35\sqrt{3}}{4}$.

260. (a) $\dfrac{\sin^3 x}{3} + C$ (b) $\dfrac{1}{3}$ (c) The function is non-negative on this interval, so the integral equals the area.

261. (a) $x = \sqrt{y}$ (b) $x = y^{\frac{1}{3}}$ (c) $x = \dfrac{1}{y}$ (d) $x = \sqrt{y-1}$

262. (a) $x = \sqrt{y-4}$ (b) $x = \dfrac{y}{3}$ (c) $x = e^y$ (d) $x = \ln y$ (e) $x = y^2$

263. $x = \dfrac{y-1}{2}$. $\displaystyle\int_1^5 \frac{y-1}{2}\,dy = \left[\frac{(y-1)^2}{4}\right]_1^5 = 4$. (A trapezium check: triangle base 2, height 4: 4.)

264. (a) $\displaystyle\int_1^4 y^{\frac{1}{2}} dy = \tfrac{2}{3}(8-1) = \tfrac{14}{3}$ (b) $\left[\tfrac{3}{4}y^{\frac{4}{3}}\right]_1^8 = \tfrac{45}{4}$ (c) $\left[\dfrac{y^3}{3}\right]_1^3 = \tfrac{26}{3}$

265. $\displaystyle\int_0^8 y^{\frac{1}{3}} dy = \left[\tfrac{3}{4}y^{\frac{4}{3}}\right]_0^8 = \tfrac{3}{4} \times 16 = 12$.

266. $\displaystyle\int_0^6 \frac{y}{2}\,dy = \left[\frac{y^2}{4}\right]_0^6 = 9$.

267. $x = \sqrt{y-1}$. $\left[\tfrac{2}{3}(y-1)^{\frac{3}{2}}\right]_2^5 = \tfrac{2}{3}(8-1) = \tfrac{14}{3}$.

268. $\left[\tfrac{2}{3}(y-3)^{\frac{3}{2}}\right]_4^7 = \tfrac{2}{3}(8-1) = \tfrac{14}{3}$.

269. $x = \dfrac{2}{y}$. $\left[2\ln y\right]_1^4 = 2\ln 4$.

270. $x = e^y$: $\displaystyle\int_0^3 e^y dy = e^3 - 1$.

271. The curve meets $y = e^2$ at $x = 2$. $\displaystyle\int_0^2 \left(e^2 - e^x\right) dx = \left[e^2 x - e^x\right]_0^2 = e^2 + 1$.

272. The part of the rectangle left of the curve is $\displaystyle\int_0^1 e^y dy = e - 1$. The wanted region is the rectangle minus this part: $e - (e-1) = 1$.

273. By dx: $\displaystyle\int_0^2 x^2 dx = \tfrac{8}{3}$. By dy: rectangle $2 \times 4 = 8$ minus $\displaystyle\int_0^4 \sqrt{y}\,dy = \tfrac{16}{3}$ gives $\tfrac{8}{3}$.

274. $x = y^{\frac{1}{3}}$ is negative for $y < 0$. $\displaystyle\int_{-8}^0 y^{\frac{1}{3}} dy = -12$ and $\displaystyle\int_0^8 = 12$. Area $= 24$.

275. For $x \le 0$, $x = -\sqrt{y}$. $\displaystyle -\int_1^9 \left(-\sqrt{y}\right) dy = \left[\tfrac{2}{3}y^{\frac{3}{2}}\right]_1^9 = \tfrac{2}{3}(27-1) = \tfrac{52}{3}$.

276. Rectangle $0 \le x \le 3, 1 \le y \le 8$ has area 21. Part below the curve: $\displaystyle\int_0^3 \left(2^x - 1\right) dx = \frac{7}{\ln 2} - 3$. Area $= 21 - \left(\dfrac{7}{\ln 2} - 3\right) = 24 - \dfrac{7}{\ln 2}$.

277. Reflecting in $y = x$ turns the region into the region between $y = 2^x$, the y-axis and the line $y = 8$ (for $0 \le x \le 3$). That is the rectangle $[0,3] \times [0,8]$ (area 24) minus the area under $y = 2^x$ from 0 to 3 (area $\frac{7}{\ln 2}$), giving $24 - \frac{7}{\ln 2}$.

278. Rectangle $[0,9] \times [0,2]$ has area 18. The area under $y = 3^x$ from 0 to 2 is $\frac{8}{\ln 3}$. So the area is $18 - \frac{8}{\ln 3}$.

279. The strips are horizontal and are stacked up the y-axis, so each strip's position is a y-value and the integral adds from the bottom y to the top y.

280. That integral uses dx with y-limits and gives the area under the curve. The correct area uses $x = \sqrt{y}$: $\int_0^4 \sqrt{y}\, dy = \frac{16}{3}$.

281. The first region is left of the curve with $\int_0^2 e^y dy = e^2 - 1$. The second is the rectangle $[0, e^2] \times [0,2]$ ($2e^2$) minus that region: $2e^2 - (e^2 - 1) = e^2 + 1$.

282. (a) Square both sides of $y = \sqrt{x}$. (b) $\int_0^2 y^2 dy = \frac{8}{3}$. (c) $\int_0^4 \sqrt{x}\, dx = \frac{16}{3}$. The rectangle $4 \times 2 = 8$ minus $\frac{8}{3}$ is $\frac{16}{3}$.

283. $\int_0^k y^2 dy = \frac{k^3}{3} = 9$, so $k^3 = 27$ and $k = 3$.

284. (a) $(x-3)(x+1) = 0$: $x = -1$ and $x = 3$. (b) $\left[\frac{x^3}{3} - x^2 - 3x\right]_{-1}^{3} = -9 - \frac{5}{3} = -\frac{32}{3}$. Area $= \frac{32}{3}$. (c) $\int_0^3 (-f(x))\, dx = 9$.

285. (a) $\log_2\big((x+1)(x-1)\big) = 3$, so $x^2 - 1 = 2^3 = 8$. (b) $x^2 = 9$, so $x = 3$ (reject $x = -3$, since $\log_2(x-1)$ is undefined).

286. (a) Quotient rule: $\frac{e^x \times x - e^x \times 1}{x^2} = \frac{e^x(x-1)}{x^2}$. (b) $x = 1$, $y = e$. $\frac{dy}{dx}$ goes from negative to positive: a minimum at $(1, e)$. (c) $\frac{e^2 - 1}{2}$.

287. (a) $h = 0.5$: 0, 0.5, 1, 1.5, 2 (b) $h = 1$: 1, 2, 3 (c) $h = 0.25$: 0, 0.25, 0.5, 0.75, 1

288. (a) $h = 1$: $0, 1, 2, 3$ (b) $h = 1$: $2, 3, 4, 5, 6$ (c) $h = 0.2$: 1, 1.2, 1.4, 1.6, 1.8, 2

289. There are n strips between a and b, so n equal widths fill the interval. There are $n + 1$ function values, one at each end of each strip.

290. (a) $\frac{2}{2}[0 + 4] = 4$ (b) $\frac{2}{2}\left[1 + \frac{1}{3}\right] = \frac{4}{3}$

291. (a) $\frac{4}{2}[0 + 2] = 4$ (b) $\frac{2}{2}[1 + 4] = 5$ (c) $\frac{1}{2}[1 + 8] = 4.5$

292. $\frac{3}{2}[3 + 9] = 18$ and $\left[x^2 + x\right]_1^4 = 20 - 2 = 18$. The graph is a straight line, so the trapezium is the region itself.

293. $h = 1$: $\frac{1}{2}\Big[3 + 2(5+6+4) + 2\Big] = 17.5$

294. $h = 3$: $\frac{3}{2}\Big[1 + 2(4+6) + 5\Big] = 39$

295. $\frac{1}{2}\Big[60 + 2(55+65+68+70) + 67\Big] = 321.5$ cf. 2020 HSC Q20

296. $h = 1$: $\frac{1}{2}\Big[1 + 2\sqrt{2} + \sqrt{5}\Big] \approx 3.03$ cf. 2022 HSC Q13

297. $h = 0.5$: $0.25\Big[1 + 2e^{0.5} + e\Big] \approx 1.754$. Exact: $e - 1 \approx 1.718$. The estimate is slightly high.

298. $h = 1$: $\frac{1}{2}\Big[1 + 2 \times \frac{1}{3} + \frac{1}{9}\Big] = \frac{8}{9}$ cf. 2025 HSC Q27(a)

299. (a) $f'' = 2 > 0$: over-estimate. (b) $f'' = -\frac{1}{4}x^{-\frac{3}{2}} < 0$: under-estimate. (c) $f'' = e^x > 0$: over-estimate.

300. $h = 2$: $\frac{2}{2}\Big[0 + 2\sqrt{2} + 2\Big] \approx 4.83$. $f'' < 0$, so too small. Exact $\frac{16}{3} \approx 5.33$.

301. $f'(x) = 2xe^{x^2}$ and $f''(x) = \left(2 + 4x^2\right)e^{x^2} > 0$. The estimate is an over-estimate. cf. 2024 HSC Q22

302. $h = 0.5$. Left $= 0.5(0 + 0.25 + 1 + 2.25) = 1.75$. Right $= 0.5(0.25 + 1 + 2.25 + 4) = 3.75$. Exact $\frac{8}{3} \approx 2.67$ lies between.

303. $h = 0.5$. Decreasing function: lower (right) $= 0.5\left[\frac{2}{3} + \frac{1}{2} + \frac{2}{5} + \frac{1}{3}\right] = 0.95$. Upper (left) $= 0.5\left[1 + \frac{2}{3} + \frac{1}{2} + \frac{2}{5}\right] \approx 1.283$. $\ln 3 \approx 1.099$ lies between.

304. $h = 0.25$. Lower (left) $= 0.25\left(1 + e^{0.25} + e^{0.5} + e^{0.75}\right) \approx 1.512$. Upper (right) $= 0.25\left(e^{0.25} + e^{0.5} + e^{0.75} + e\right) \approx 1.942$. Exact $e - 1 \approx 1.718$.

305. The two trapezia have areas $\frac{h}{2}\left[f(x_0) + f(x_1)\right]$ and $\frac{h}{2}\left[f(x_1) + f(x_2)\right]$. Adding doubles the middle value.

306. $n = 1$: 4. $n = 2$: $\frac{1}{2}[0 + 2 + 4] = 3$. The exact value is $\frac{8}{3} \approx 2.67$. The estimates fall toward it, so more strips means higher precision.

307. (a) $\frac{8}{9}$ (b) $\left[-\frac{3^{-x}}{\ln 3}\right]_0^2 = \frac{1}{\ln 3} - \frac{1}{9\ln 3} = \frac{8}{9\ln 3}$. (c) $f''(x) = (\ln 3)^2 3^{-x} > 0$, so the curve is concave up and the chords lie above it. cf. 2025 HSC Q27

308. B. $h = 2$: $\frac{2}{2}\left[1 + 2(5) + 3\right] = 14$.

309. (a) $x = \pm 2$. (b) $\int_{-2}^{2}\left(3x^2 - 12\right)dx = \Big[x^3 - 12x\Big]_{-2}^{2} = -32$. Area $= 32$. (c) $\Big[x^3 - 12x\Big]_2^3 = (27 - 36) - (8 - 24) = 7$.

310. (a) $e^{-x} - xe^{-x} = e^{-x}(1 - x)$. (b) $\frac{x-1}{e^x} = -e^{-x}(1 - x)$, so the integral is $-\Big[xe^{-x}\Big]_0^1 = -\frac{1}{e}$.

311. (a) $\dfrac{T_{n+1}}{T_n} = \frac{1}{2}$, a constant. (b) $\dfrac{5}{1-\frac{1}{2}} = 10$. (c) $5\left(\frac{1}{2}\right)^{n-1} < 0.01$, so $\left(\frac{1}{2}\right)^{n-1} < 0.002$ and $n-1 > \dfrac{\ln 500}{\ln 2} \approx 8.97$. So $n = 10$.

312. (a) $r = \frac{1}{2}$: 2 (b) $r = \frac{1}{4}$: $\frac{4}{3}$ (c) $r = \frac{1}{10}$: $\frac{10}{9}$

313. $a = 1$, $r = \dfrac{1}{b}$ with $|r| < 1$: $\dfrac{1}{1-\frac{1}{b}} = \dfrac{b}{b-1}$.

314. The ratio is $\frac{1}{b}$ and a limiting sum needs $|r| < 1$.

315. $\left[-\dfrac{3^{-x}}{\ln 3}\right]_0^2 = \dfrac{1}{\ln 3} - \dfrac{1}{9\ln 3} = \dfrac{8}{9\ln 3}$.

316. $\dfrac{1-\frac{1}{8}}{\ln 2} = \dfrac{7}{8\ln 2}$ and $\dfrac{1-\frac{1}{5}}{\ln 5} = \dfrac{4}{5\ln 5}$.

317. $\dfrac{1-\frac{1}{16}}{\ln 2} = \dfrac{15}{16\ln 2}$.

318. (a) $32\ln 2 > 15$, so $2^{32} > e^{15}$ (b) $3^{27} > e^{16}$ (c) $5^4 < e^7$

319. (a) $2^8 > e^3$ (b) $4^{16} > e^7$ (c) $3^3 < e^4$ (d) $7^4 > e^5$

320. $2^8 = 256 > e^3 \approx 20.1$. $3^{27} \approx 7.6 \times 10^{12} > e^{16} \approx 8.9 \times 10^6$.

321. (a) $a = 1$, $r = \frac{1}{3}$: $\dfrac{1}{1-\frac{1}{3}} = \dfrac{3}{2}$. (b) $\dfrac{1-\frac{1}{3}}{\ln 3} = \dfrac{2}{3\ln 3}$. (c) $\dfrac{2}{3\ln 3} < \dfrac{3}{2}$, so $4 < 9\ln 3$ and $\ln 3 > \frac{4}{9}$. Hence $3^9 > e^4$.

322. Limiting sum $\frac{4}{3}$. $\displaystyle\int_0^3 4^{-x}dx = \frac{63}{64\ln 4}$. So $\dfrac{63}{64\ln 4} < \dfrac{4}{3}$, giving $189 < 256\ln 4$ and $4^{256} > e^{189}$.

323. Limiting sum $\frac{10}{9}$. $\displaystyle\int_0^1 10^{-x}dx = \frac{9}{10\ln 10}$. So $\dfrac{9}{10\ln 10} < \dfrac{10}{9}$, giving $81 < 100\ln 10$ and $10^{100} > e^{81}$.

324. $\displaystyle\int_1^4 \frac{1}{x}\,dx = \ln 4$ and $\dfrac{1}{x}$ is decreasing. Right-end: $\frac{1}{2}+\frac{1}{3}+\frac{1}{4} = \frac{13}{12} < \ln 4$. Left-end: $\ln 4 < 1+\frac{1}{2}+\frac{1}{3} = \frac{11}{6}$.

325. Right-end: $\frac{1}{2}+\frac{1}{3} = \frac{5}{6} < \ln 3$. Left-end: $1+\frac{1}{2} = \frac{3}{2} > \ln 3$.

326. Right-end: $\frac{1}{2}+\frac{1}{3}+\frac{1}{4}+\frac{1}{5} = \frac{77}{60}$. Left-end: $1+\frac{1}{2}+\frac{1}{3}+\frac{1}{4} = \frac{25}{12}$. From $\ln 5 > \frac{77}{60}$, $60\ln 5 > 77$ and $5^{60} > e^{77}$.

327. $\dfrac{2}{2}\left[1+\frac{1}{3}\right] = \frac{4}{3}$. $f''(x) = \dfrac{2}{x^3} > 0$, so the curve is concave up and the estimate is too large. Hence $\ln 3 < \frac{4}{3}$ and $3^3 < e^4$.

328. $\dfrac{1}{2}\left[1+\frac{1}{2}\right] = \frac{3}{4} > \ln 2$, so $4\ln 2 < 3$ and $2^4 < e^3$.

329. $h = 1$: $\dfrac{1}{2}\left[1+\frac{2}{2}+\frac{1}{3}\right] = \frac{7}{6} > \ln 3$. So $6\ln 3 < 7$ and $3^6 < e^7$.

330. On $[k, k+1]$, $f(x) \leq f(k)$ because f is decreasing. So $\int_k^{k+1} f(x)\,dx < f(k) \times 1$. Adding for $k = 0, 1, \ldots, n-1$ gives the result.

331. e^x is an increasing function, so applying it to both sides keeps the inequality and $e^{\ln A} = A$.

332. $\dfrac{15}{16\ln 2} < 2$ gives $\ln 2 > \frac{15}{32} \approx 0.47$, so $2^{32} > e^{15}$. To prove $e^{20} < 2^{29}$ you need $\ln 2 > \frac{20}{29} \approx 0.69$, which this estimate is too weak to give.

333. (a) $\dfrac{1}{1-\frac{1}{7}} = \dfrac{7}{6}$. (b) $\dfrac{1-\frac{1}{7}}{\ln 7} = \dfrac{6}{7\ln 7}$. (c) $\dfrac{6}{7\ln 7} < \dfrac{7}{6}$, so $36 < 49\ln 7$ and $7^{49} > e^{36}$. cf. 2022 HSC Q29

334. A. The function is decreasing, so right-end sums are lower bounds and left-end sums are upper bounds. cf. 2025 HSC Q27(c)

335. (a) $\Big[\ln x\Big]_1^e = 1$. (b) $\dfrac{e-1}{2}\left(1+\dfrac{1}{e}\right) \approx 1.175$. (c) $\dfrac{1}{x}$ is concave up ($f'' = \dfrac{2}{x^3} > 0$), so the chord lies above the curve.

336. (a) $y = \ln\left(x^2+1\right) + \frac{1}{2}\sin 2x + C$. $3 = C$, so $y = \ln\left(x^2+1\right) + \frac{1}{2}\sin 2x + 3$. (b) $y = \ln\left(\frac{\pi^2}{16}+1\right) + \frac{1}{2} + 3$.

337. (a) $\dfrac{\binom{4}{2}}{16} = \dfrac{3}{8}$ (b) $1 - \dfrac{1}{16} = \dfrac{15}{16}$ (c) $\dfrac{1}{2} \times \dfrac{1}{2} = \dfrac{1}{4}$